

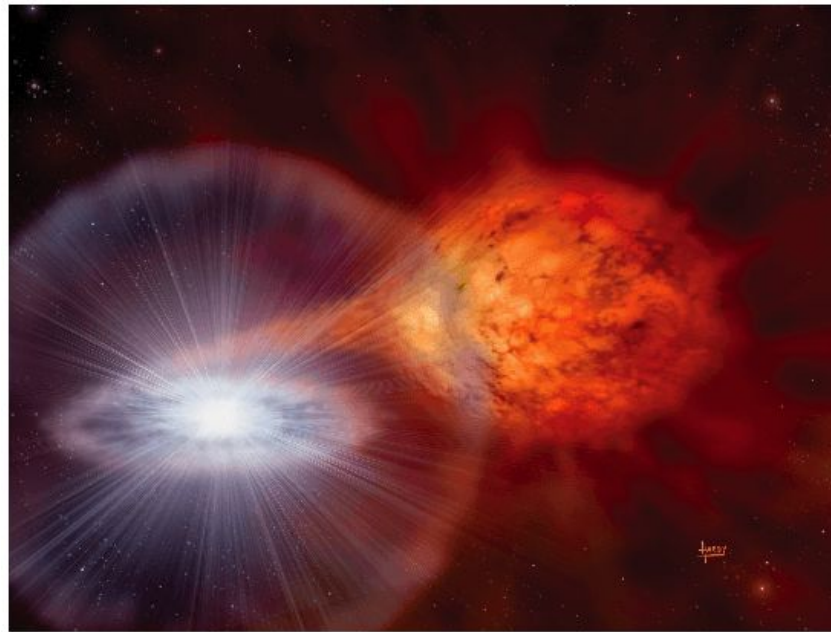
# Simulation-based Inference: Connecting Supernova Cosmology with Astrophysics

Matt Grayling



# Talk Overview

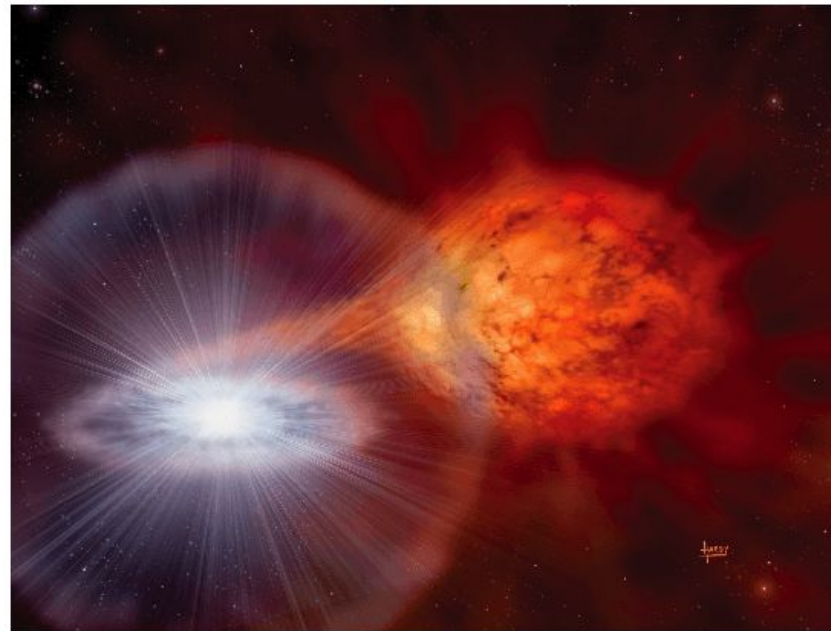
- 1) Introduction to type Ia supernovae and supernova cosmology
- 2) Investigating the 'mass step' of type Ia supernovae
- 3) SBI for supernova cosmology



Credit: David A. Hardy, © David A. Hardy/astroart.org

# Talk Overview

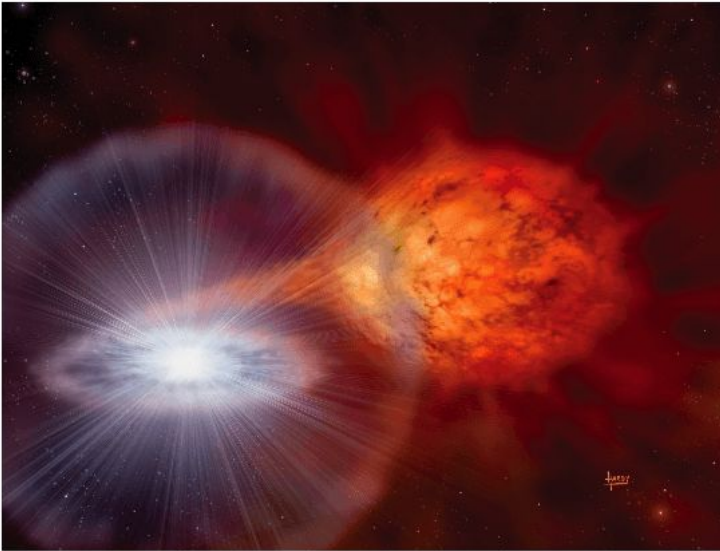
- 1) Introduction to type Ia supernovae and supernova cosmology
- 2) Investigating the 'mass step' of type Ia supernovae
- 3) SBI for supernova cosmology



Credit: David A. Hardy, © David A. Hardy/astroart.org

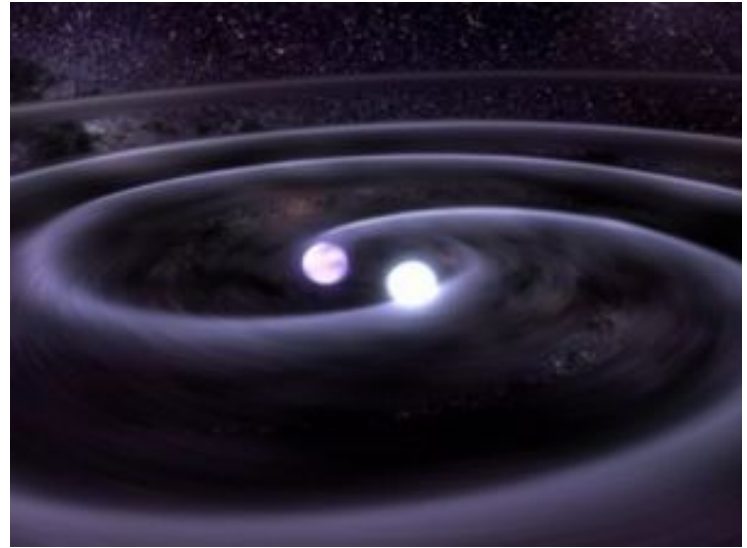
# Type Ia Supernovae

## Single Degenerate



Credit: David A. Hardy, © David A. Hardy/astroasrt.org

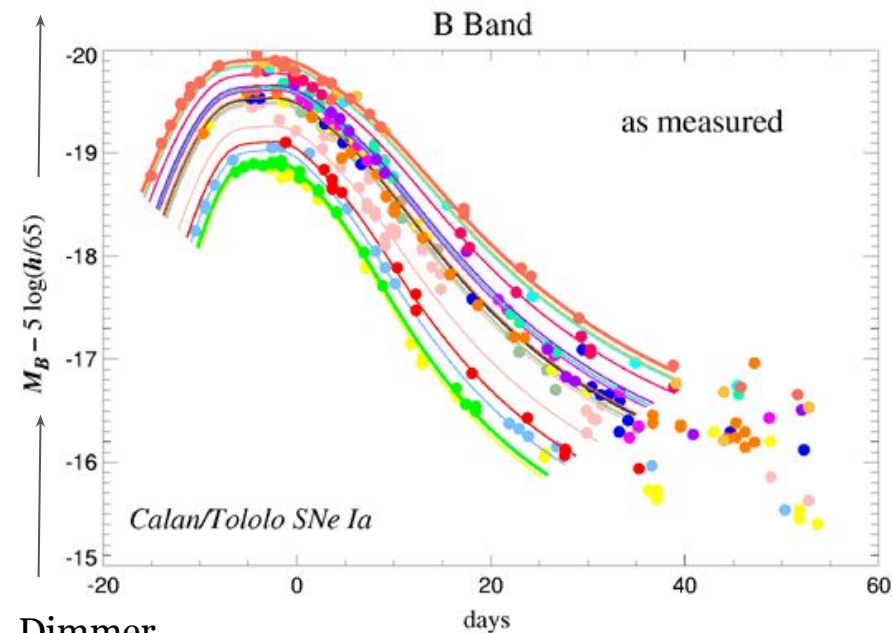
## Double Degenerate



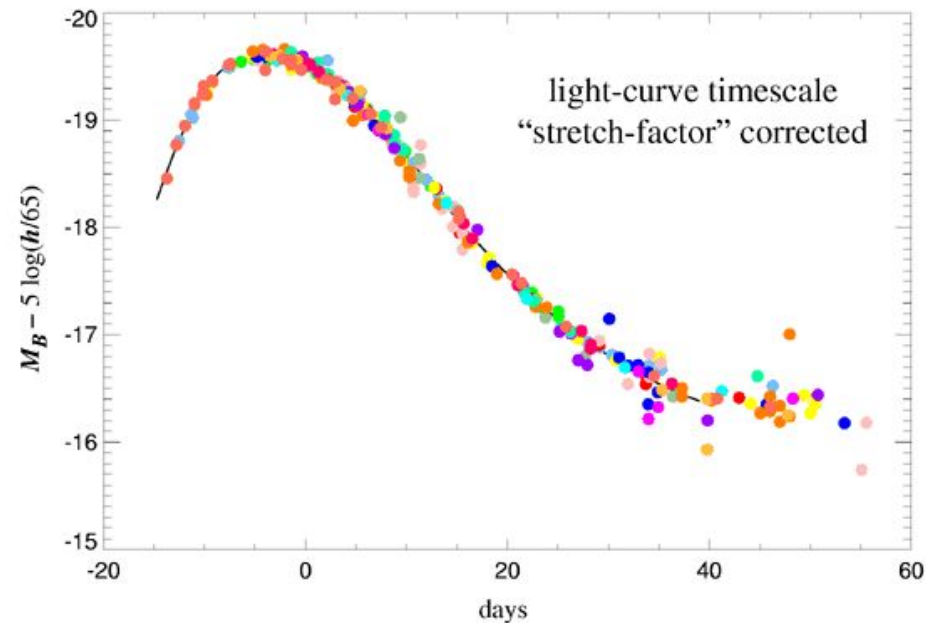
Copyright: GSFC/D. Berry.

# Standard(isable) Candles

Brighter



Dimmer

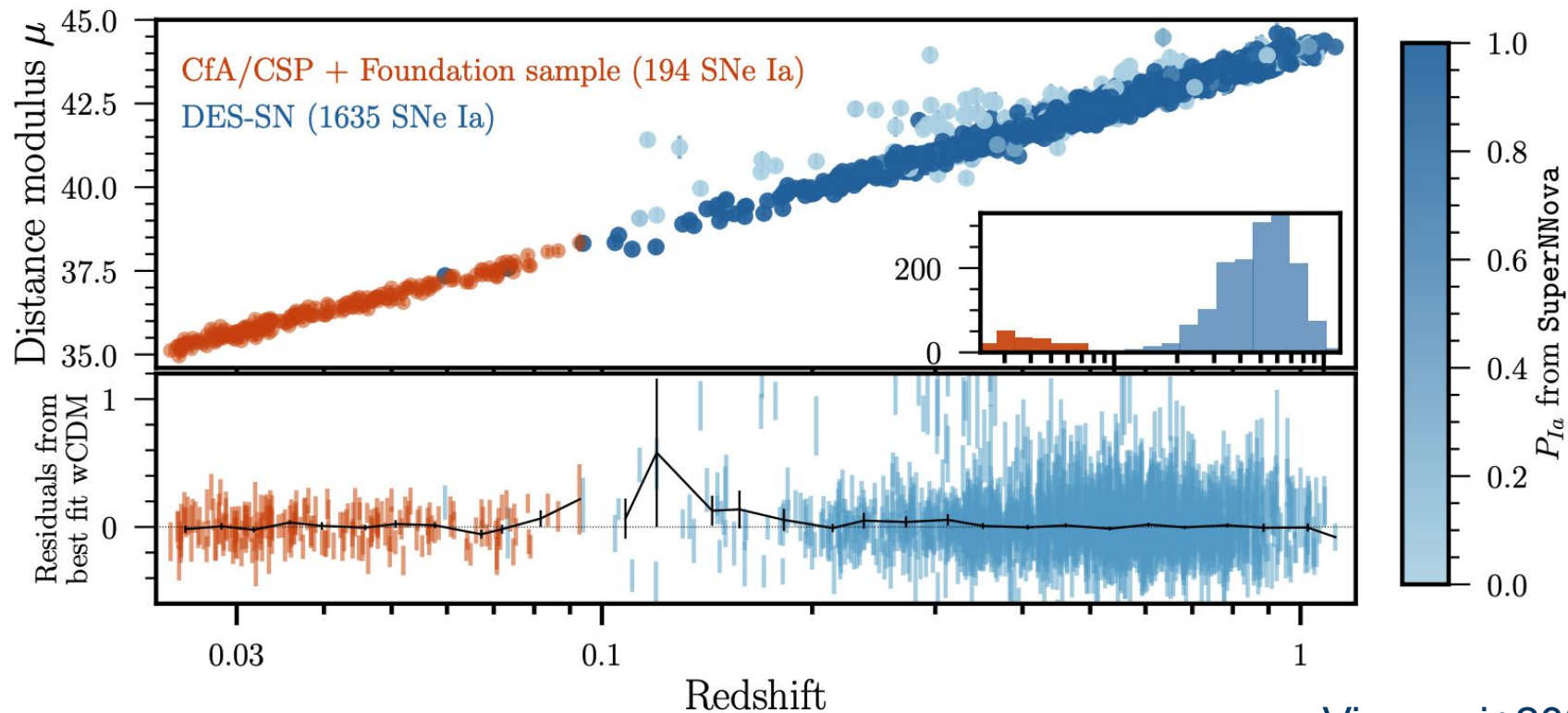


*From Supernova Cosmology Project*

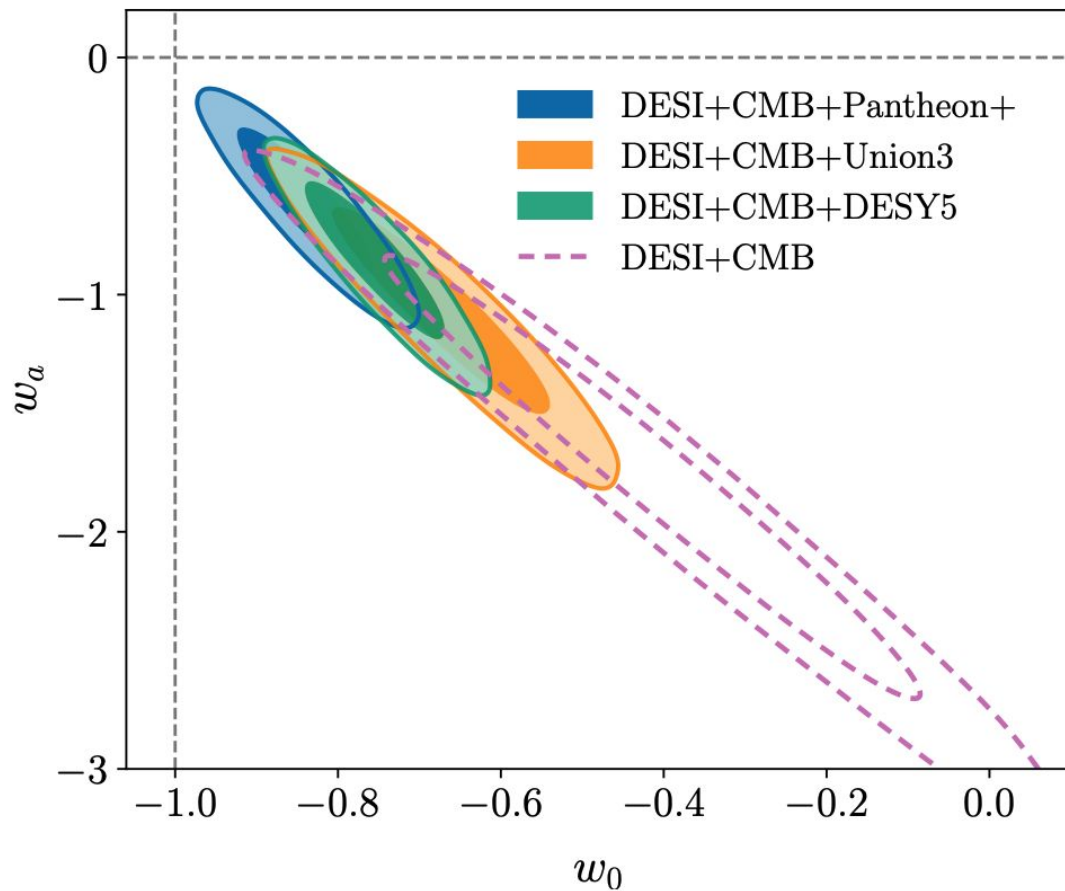
## Calculating Distance

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

# The Hubble Diagram



# Crisis in Cosmology?



DESI Collaboration  
2025



Supernova cosmology is easy, right?

# The Impact of Dust

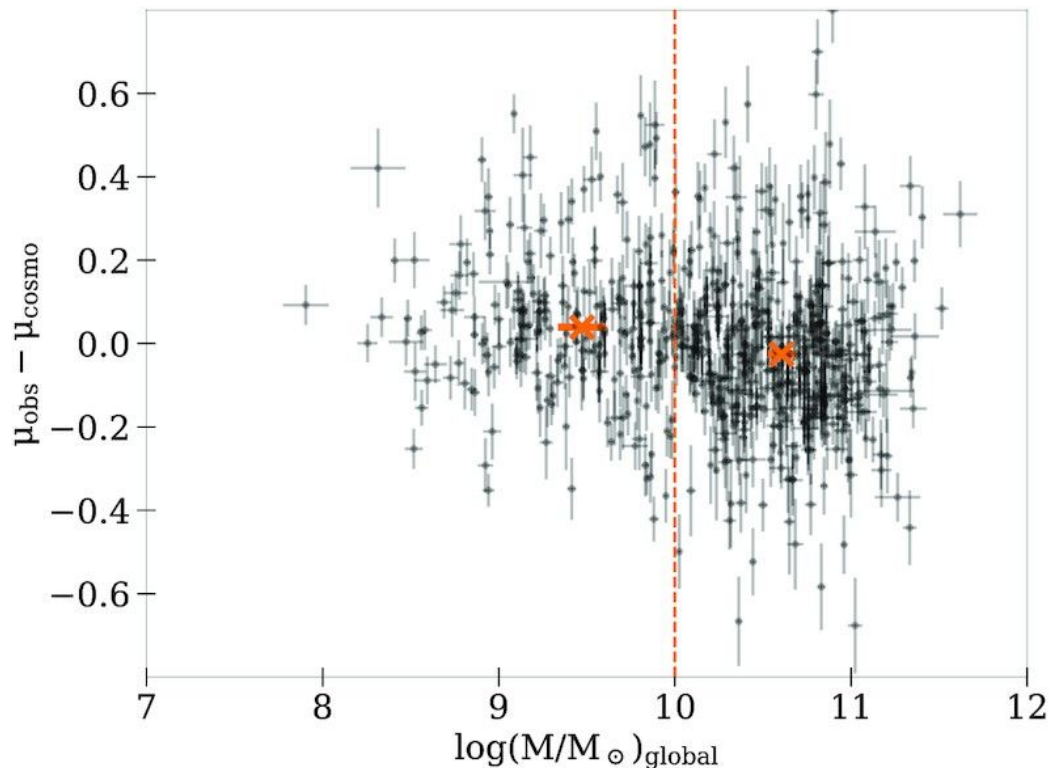
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

$$c^s = c_{\text{int}}^s + E(B - V)^s$$

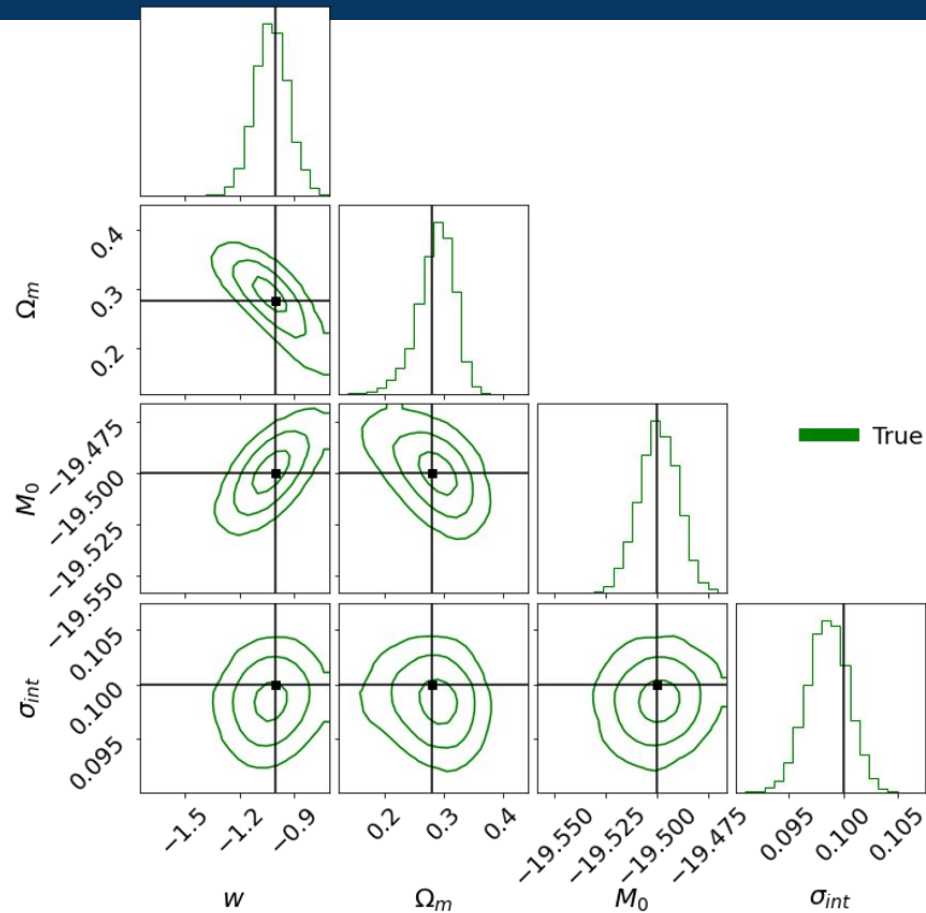
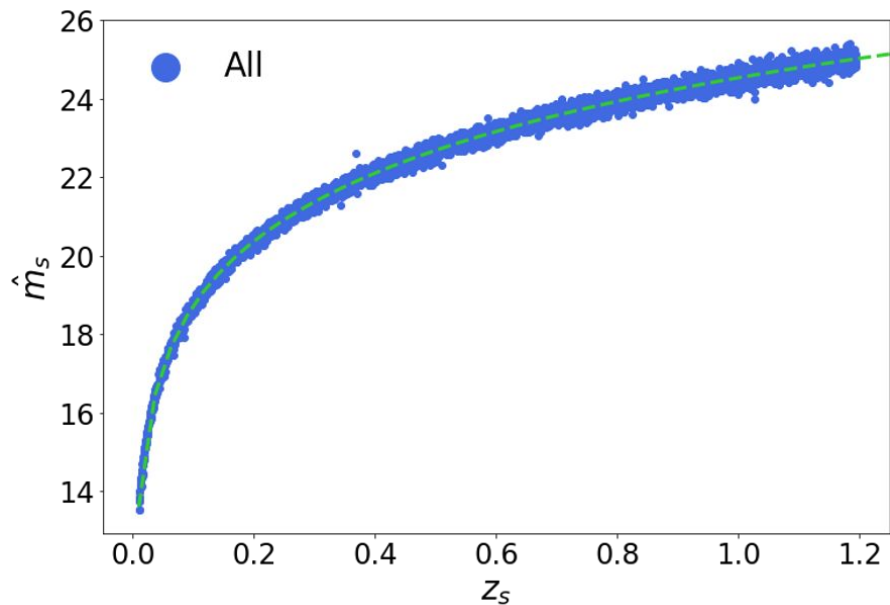

- Colour - one parameter for two separate effects
- Correctly handling dust is key to uncover intrinsic SN properties and accurately estimate distances

# The Mass Step

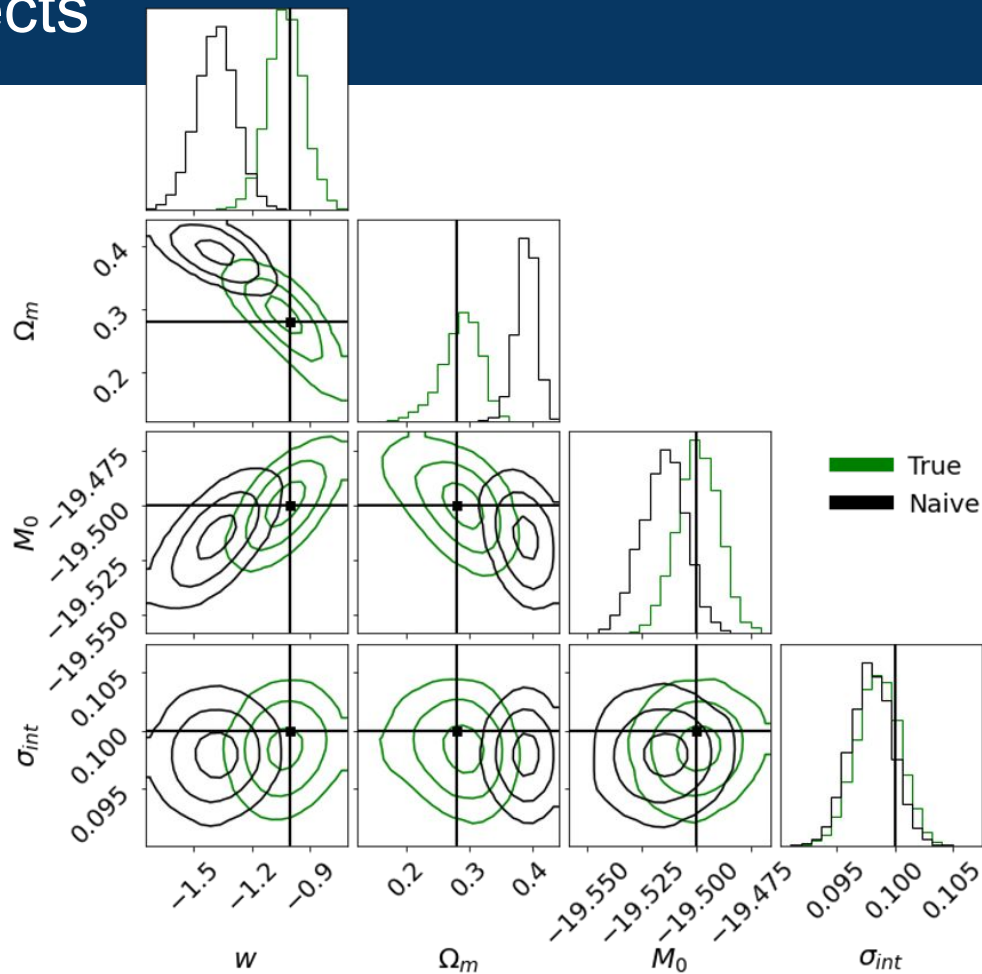
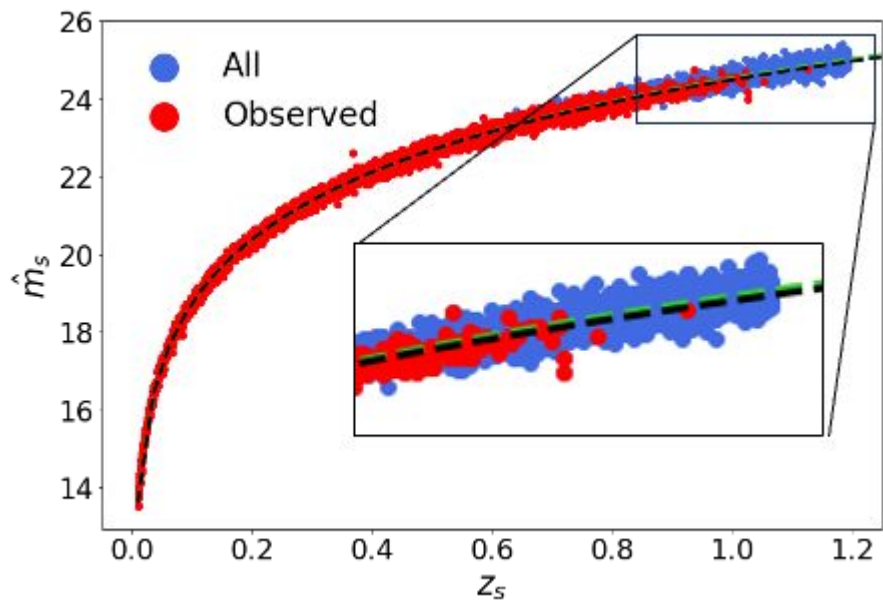
- SN Ia properties depend on environment
- Just different dust properties?
  - SN populations identical, just appear different?
- Or intrinsic differences?
  - Interesting physical implications, different progenitor channels?



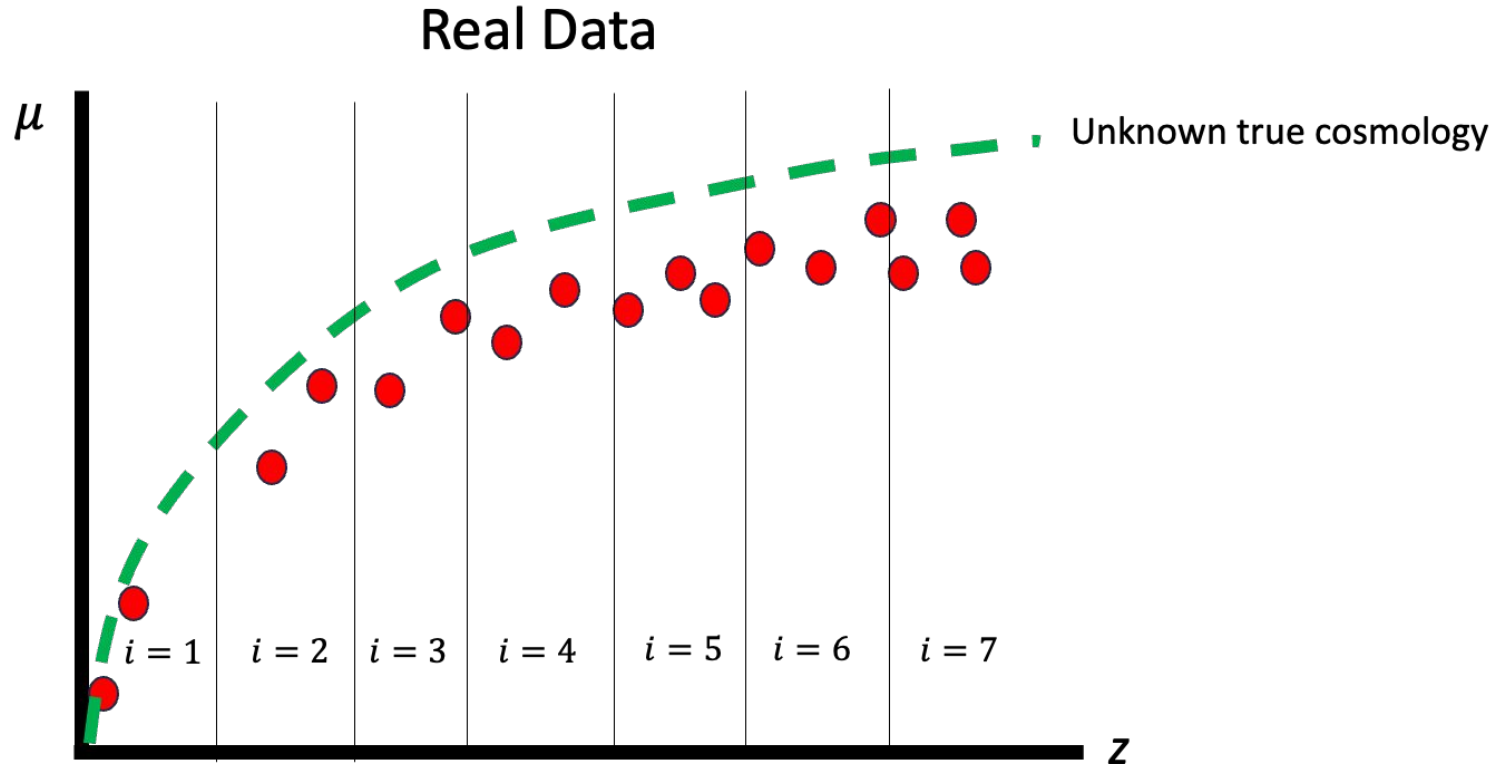
# The Impact of Selection Effects



# The Impact of Selection Effects

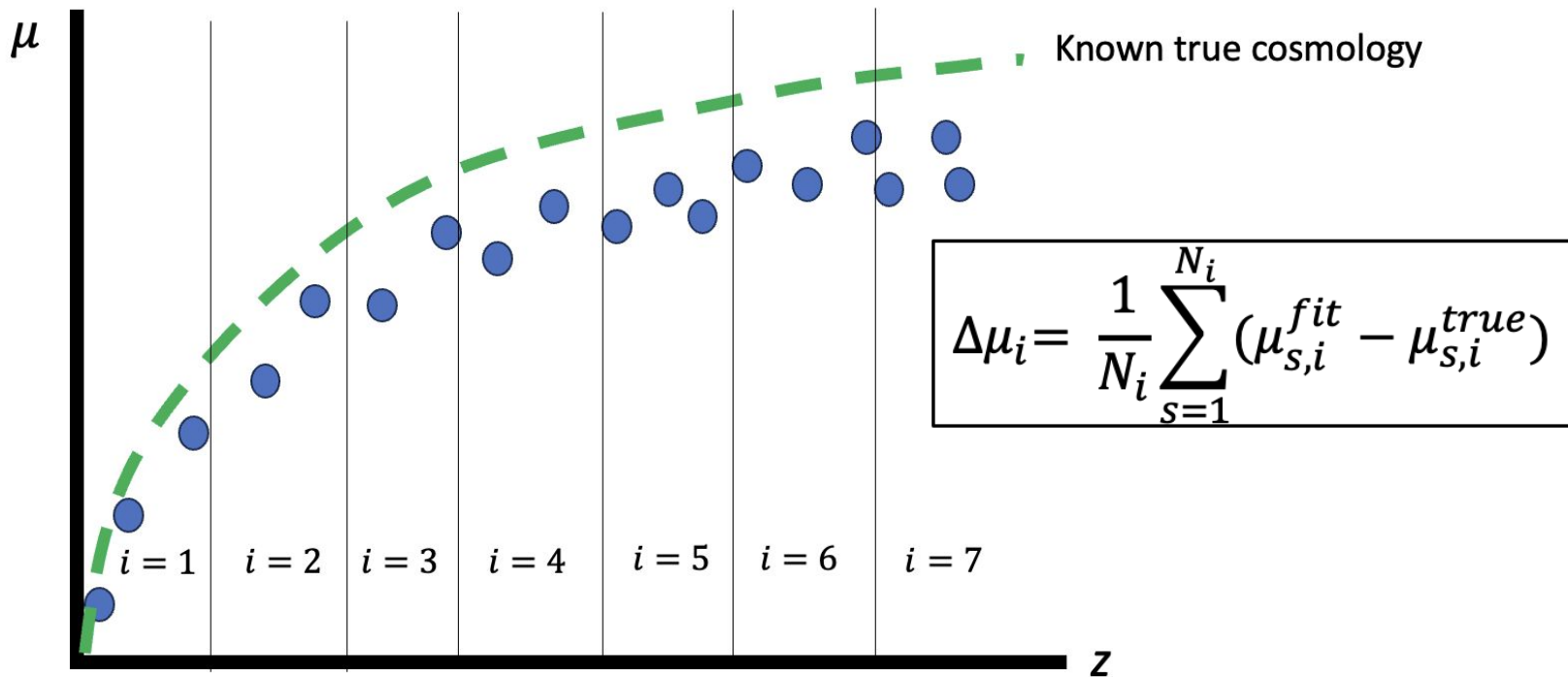


# The Impact of Selection Effects

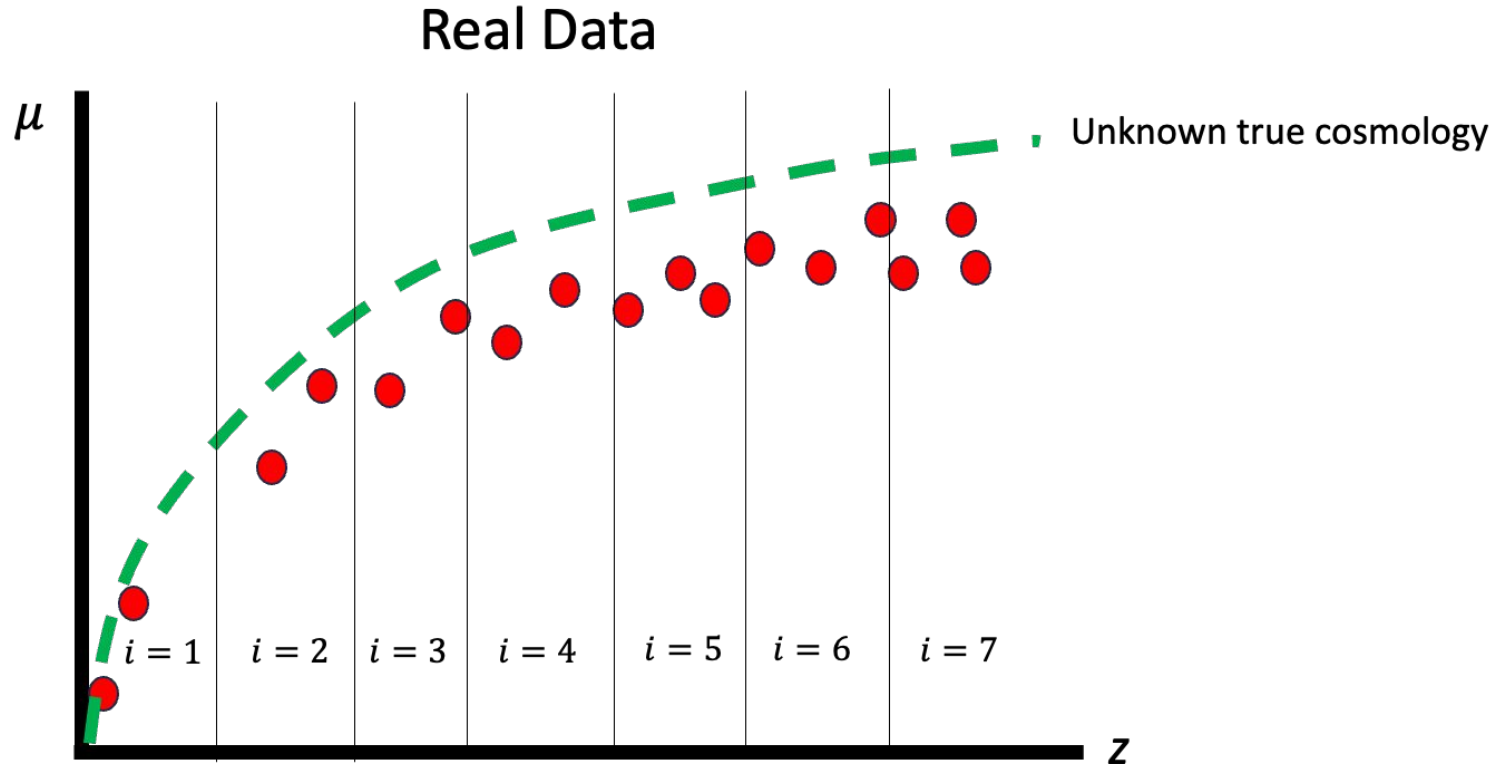


# The Impact of Selection Effects

## SNANA Simulations

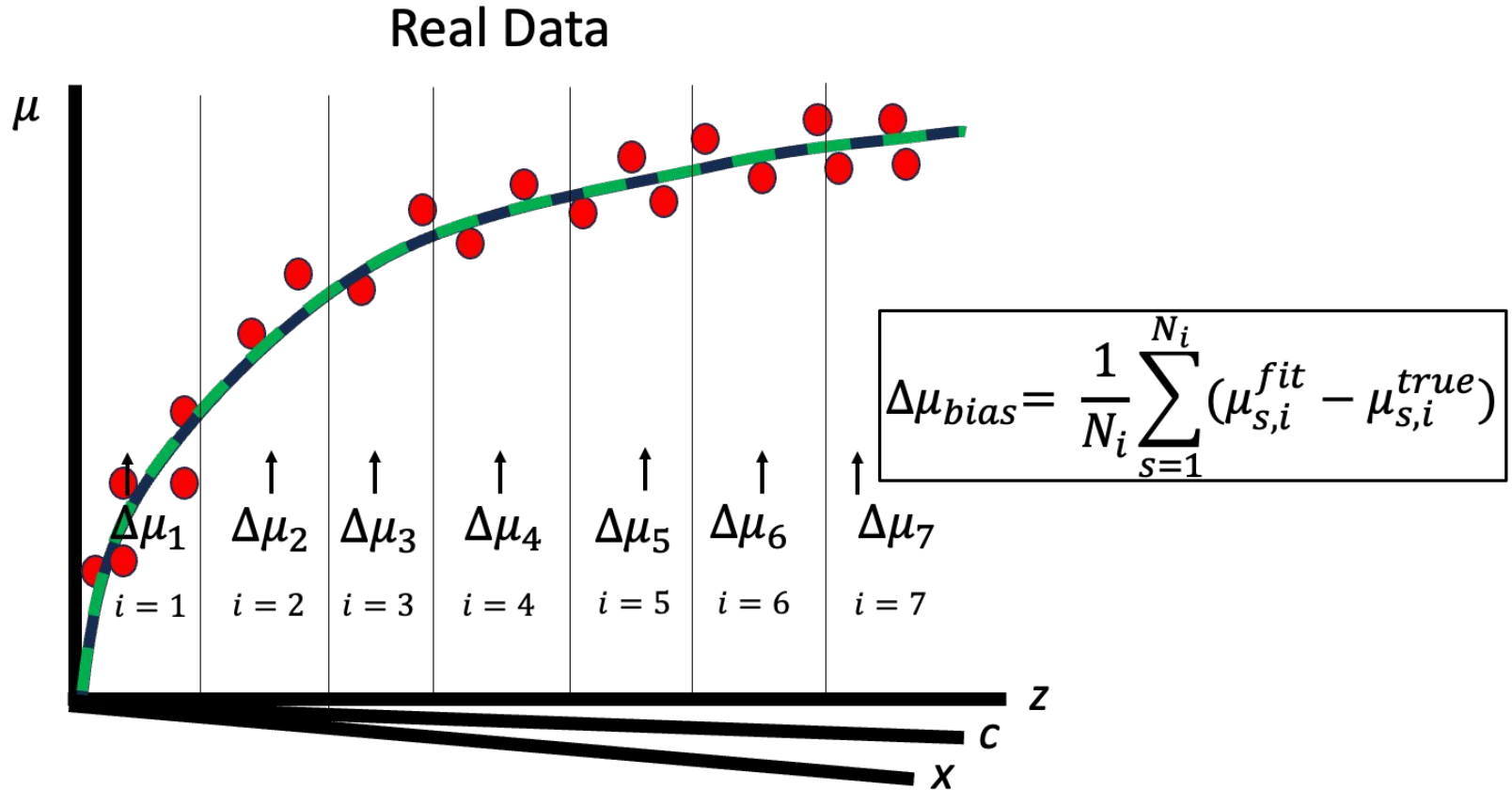


# The Impact of Selection Effects

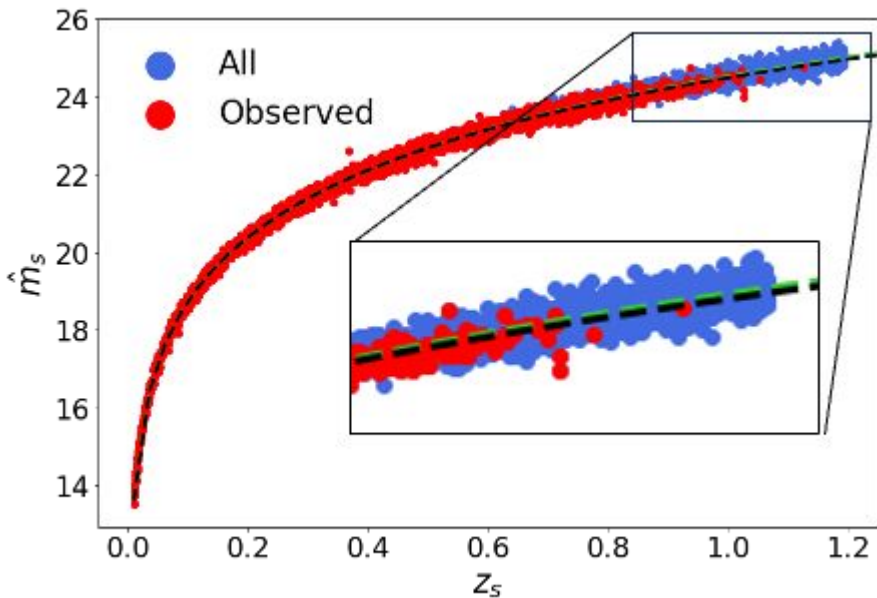




# The Impact of Selection Effects



# Correcting for Selection Effects



- We typically correct for selection effects through simulations
- This requires a fiducial cosmological model, and a fiducial model for the population of SNe Ia
- To be confident in our results, these fiducial models need to be correct

# Challenges in Supernova Cosmology

The majority of systematic uncertainty in DES5YR relates to unknown astrophysics

## Systematic $w$ uncertainty budget

DES-SN5YR, no CMB prior

$\sigma_w^{\text{SN-only (Total): 0.152}$

$\sigma_w^{\text{SN-only (stat) 0.132}$

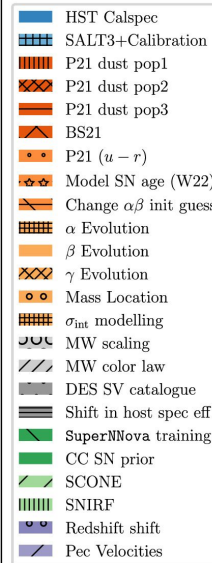
$\sigma_w^{\text{SN-only (syst-only) 0.076}$

DES-SN5YR + CMB prior

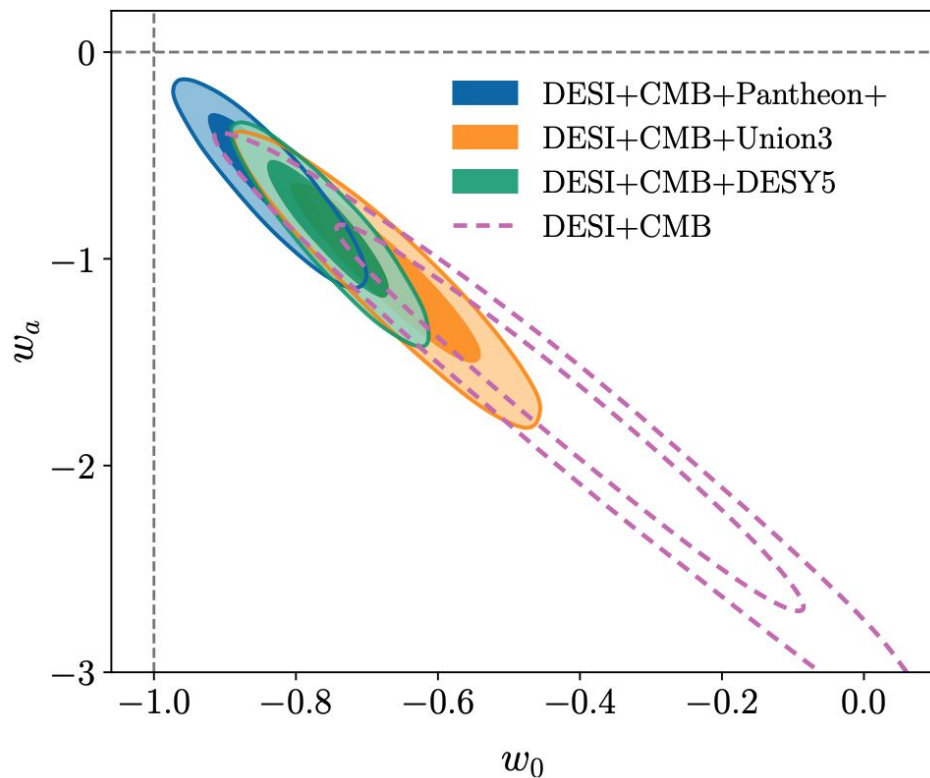
$\sigma_w^{\text{SN+CMB (Total): 0.036}$

$\sigma_w^{\text{SN+CMB (stat) 0.032}$

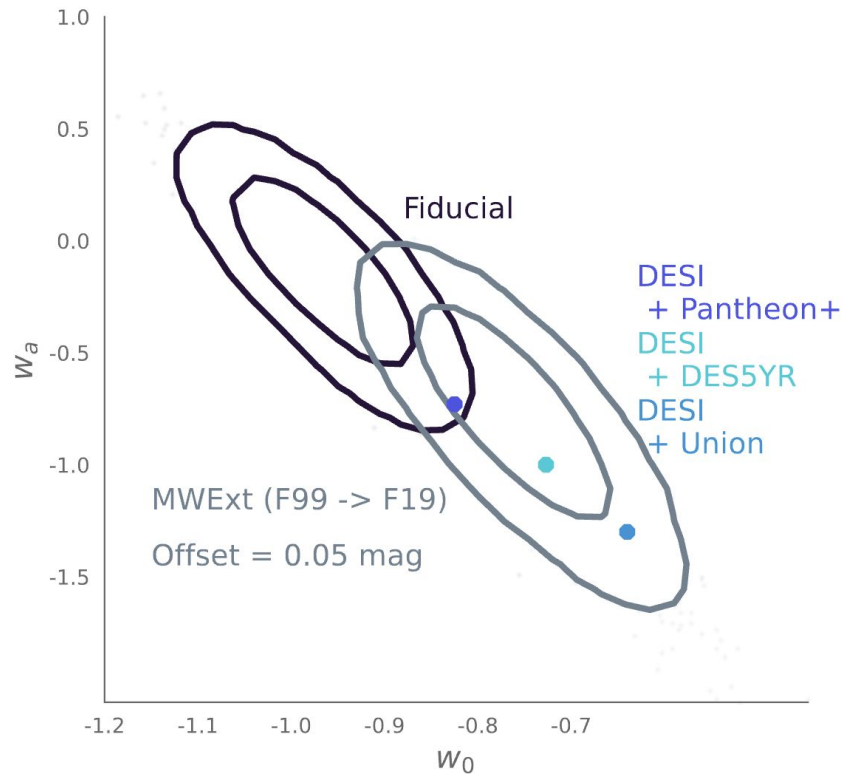
$\sigma_w^{\text{SN+CMB (syst-only) 0.017}$



# Crisis in Cosmology?



DESI Collaboration 2025

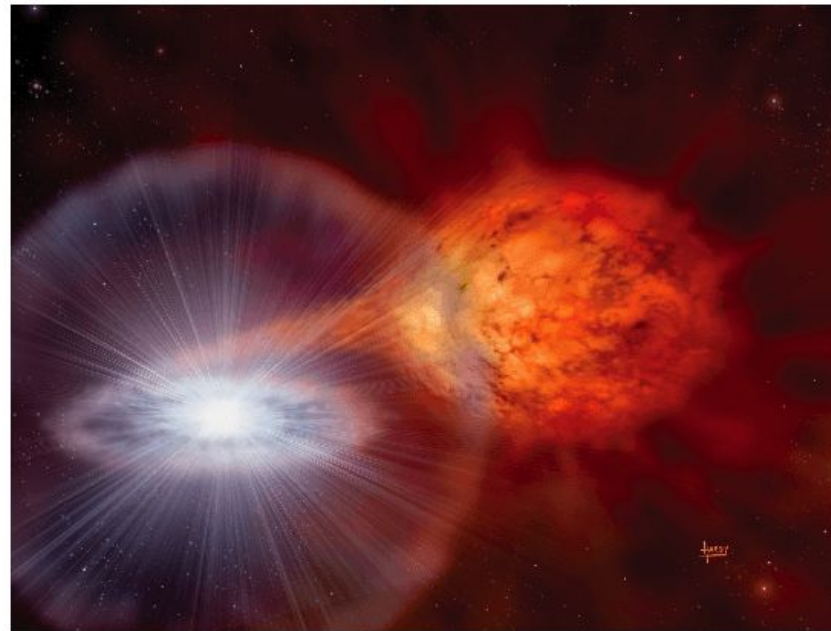


Dhawan+24

Supernova cosmology is inseparably  
linked to supernova astrophysics

# Talk Overview

- 1) Introduction to type Ia supernovae and supernova cosmology
- 2) Investigating the 'mass step' of type Ia supernovae
- 3) SBI for supernova cosmology



Credit: David A. Hardy, © David A. Hardy/astroart.org

# Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

+

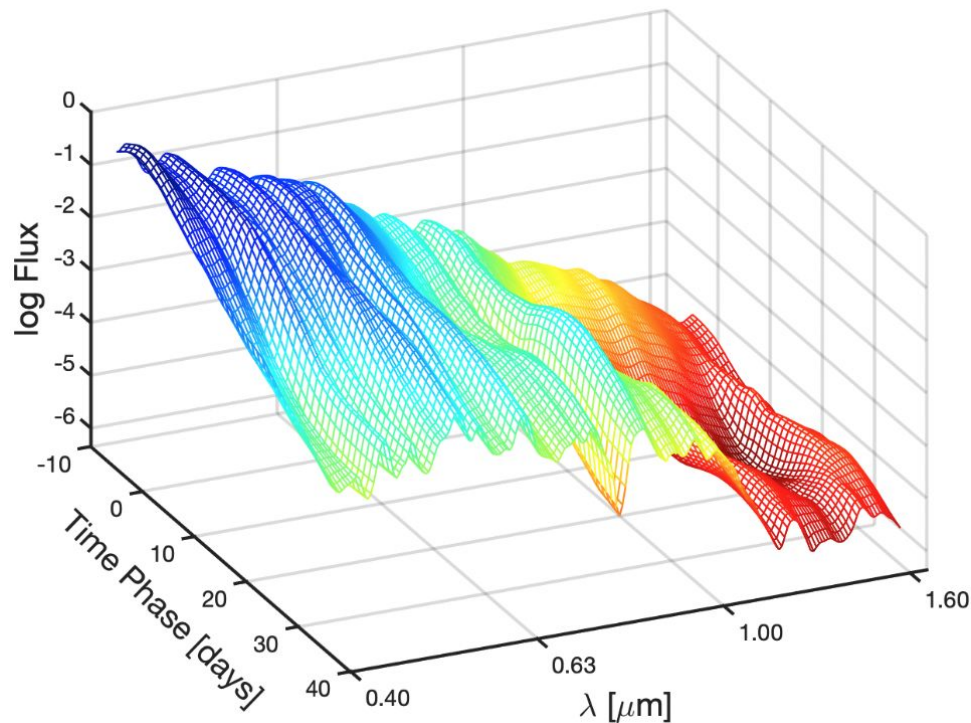
+

+



# Recipe for a Type Ia Supernova Model

Type Ia SN SED  
=  
**Mean Intrinsic SED**  
+  
+  
+

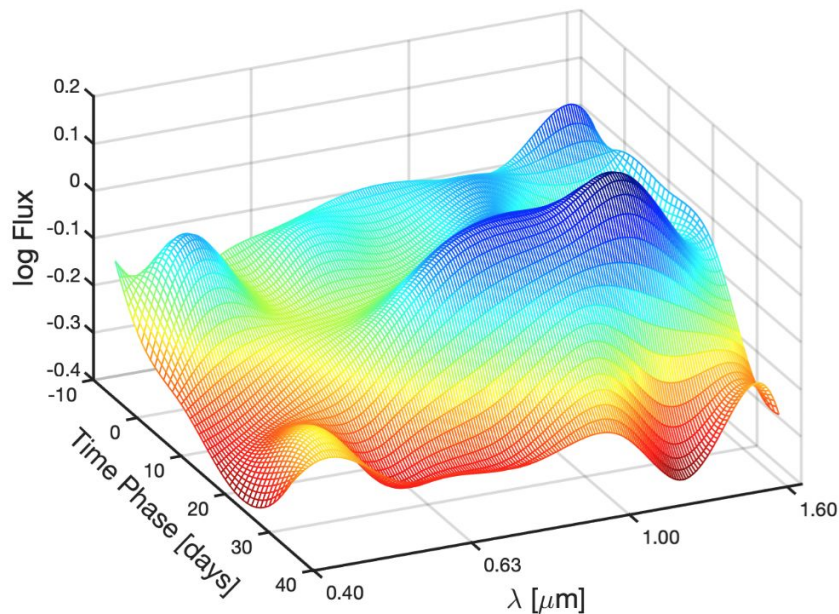




# Recipe for a Type Ia Supernova Model

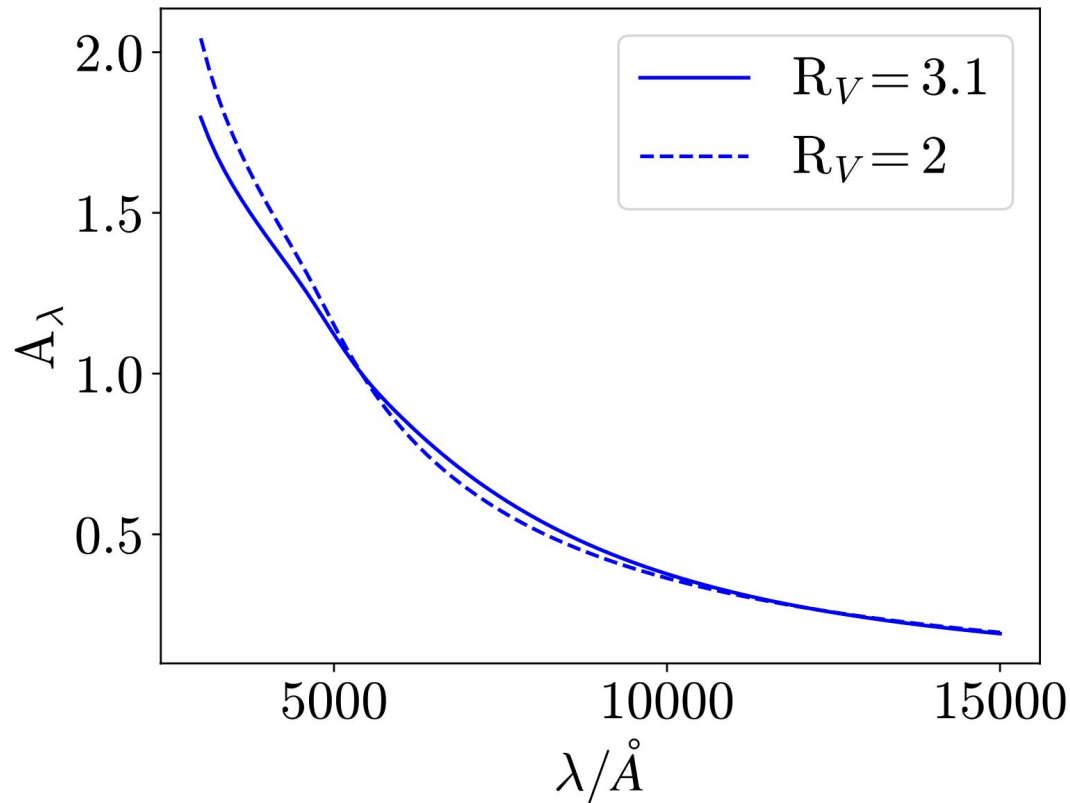
Type Ia SN SED  
=  
Mean Intrinsic SED  
+  
**Effect of Stretch**  
+  
+

$\theta_1 \times$



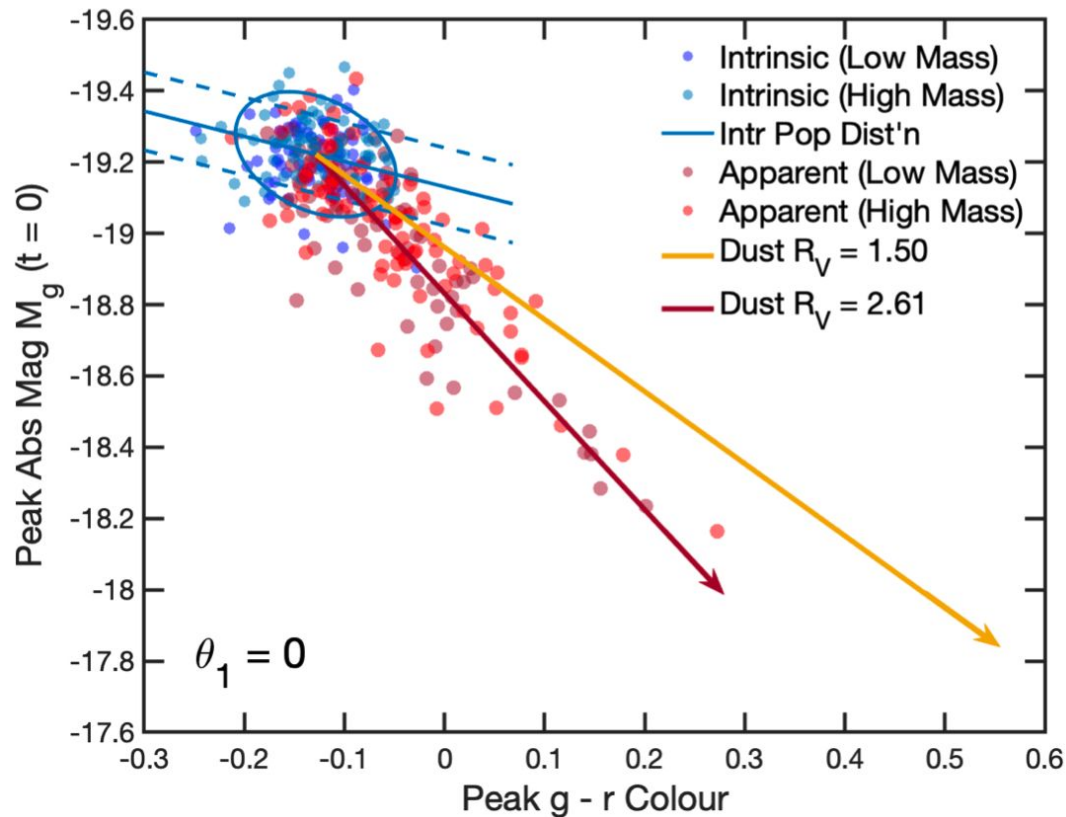
# Recipe for a Type Ia Supernova Model

Type Ia SN SED  
=  
Mean Intrinsic SED  
+  
Effect of Stretch  
+  
**Dust Extinction**  
+

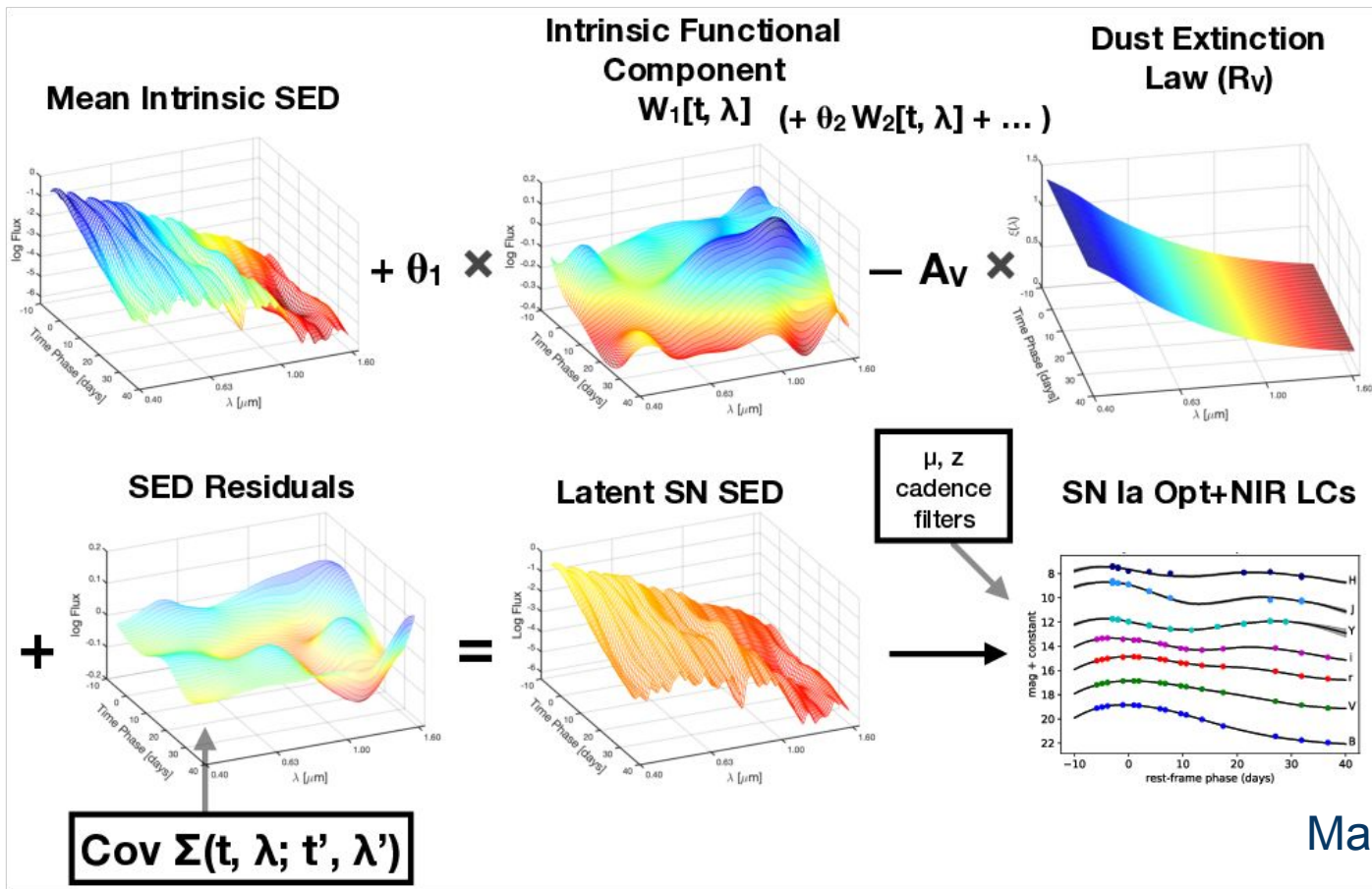


# Recipe for a Type Ia Supernova Model

Type Ia SN SED  
=  
Mean Intrinsic SED  
+  
Effect of Stretch  
+  
Dust Extinction  
+  
**Intrinsic Colour**

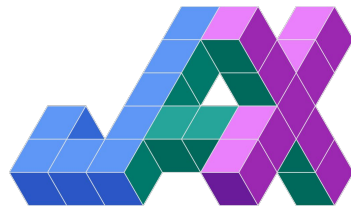


# The BayeSN Model



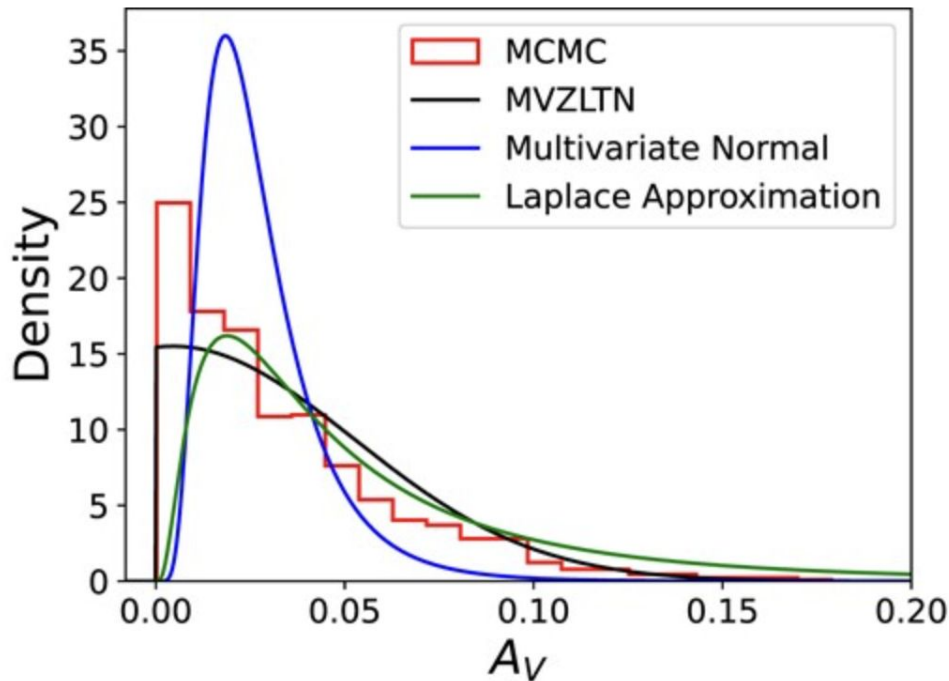
# Scaling BayeSN

- BayeSN is computationally expensive...
- ...but we have been able to achieve 100x speed ups using numpyro/jax and running on GPU
  - Very fast HMC
- Inferences that used to take days now take minutes

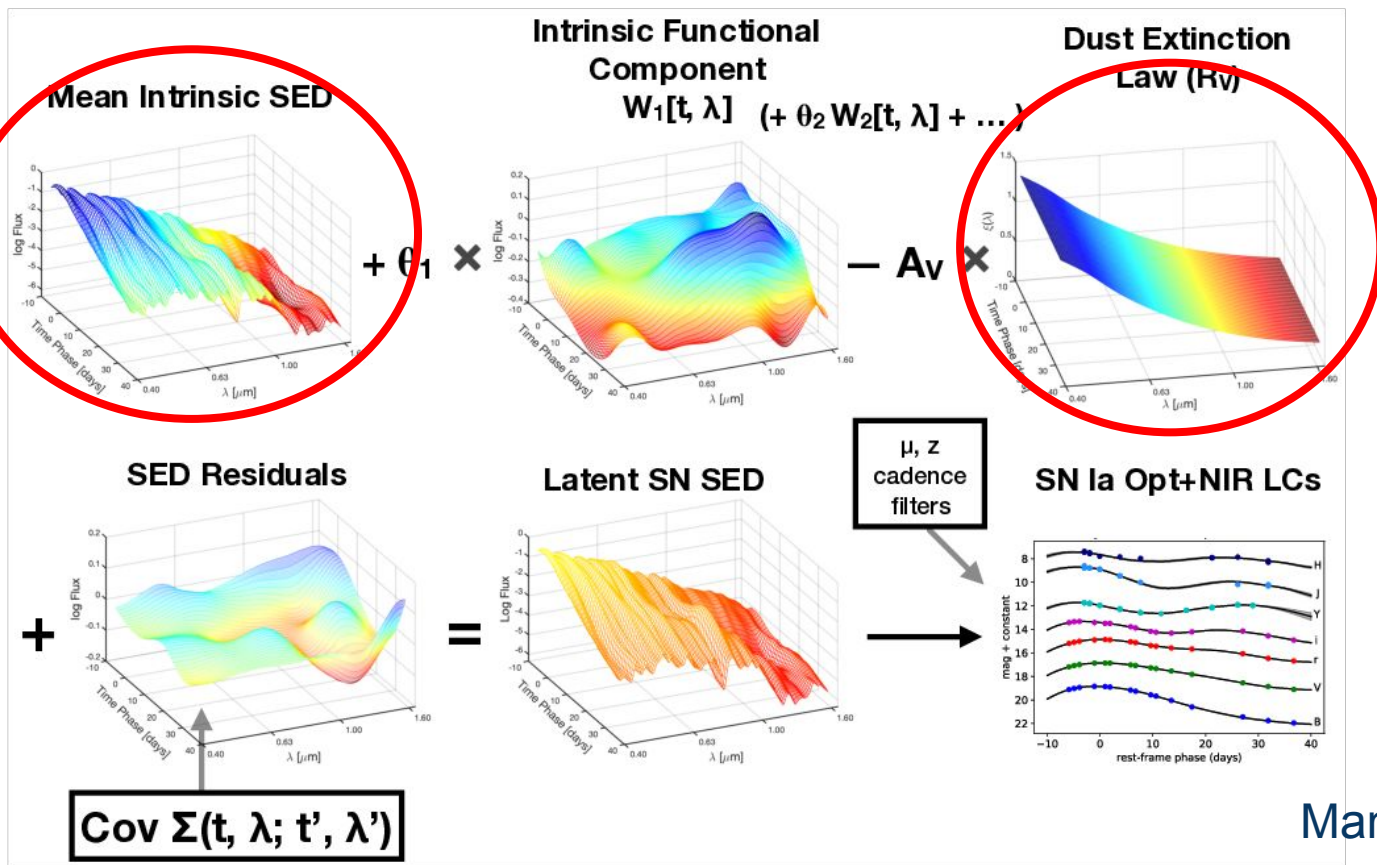


# Scaling BayeSN

- Ana Sofia Uzsoy developed variational inference methods for BayeSN
- Created custom method to account for truncated distributions, allows VI to more closely match MCMC posteriors
- Also runs on GPU, further performance gains!



# Investigating the Mass Step

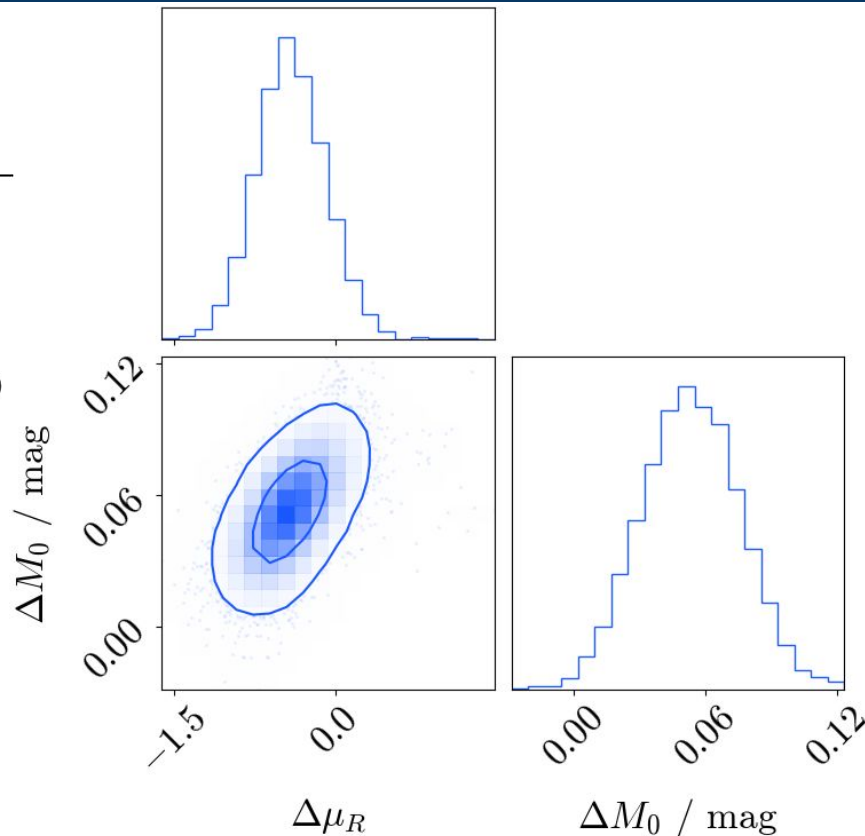


# Cause of the Mass Step

| Sample    | Intrinsic Mass Step / mag |
|-----------|---------------------------|
| Combined* | $0.049 \pm 0.019$         |
| DES5YR    | $0.053 \pm 0.022$         |

\*Combined = DES3YR+Foundation+PS1MD

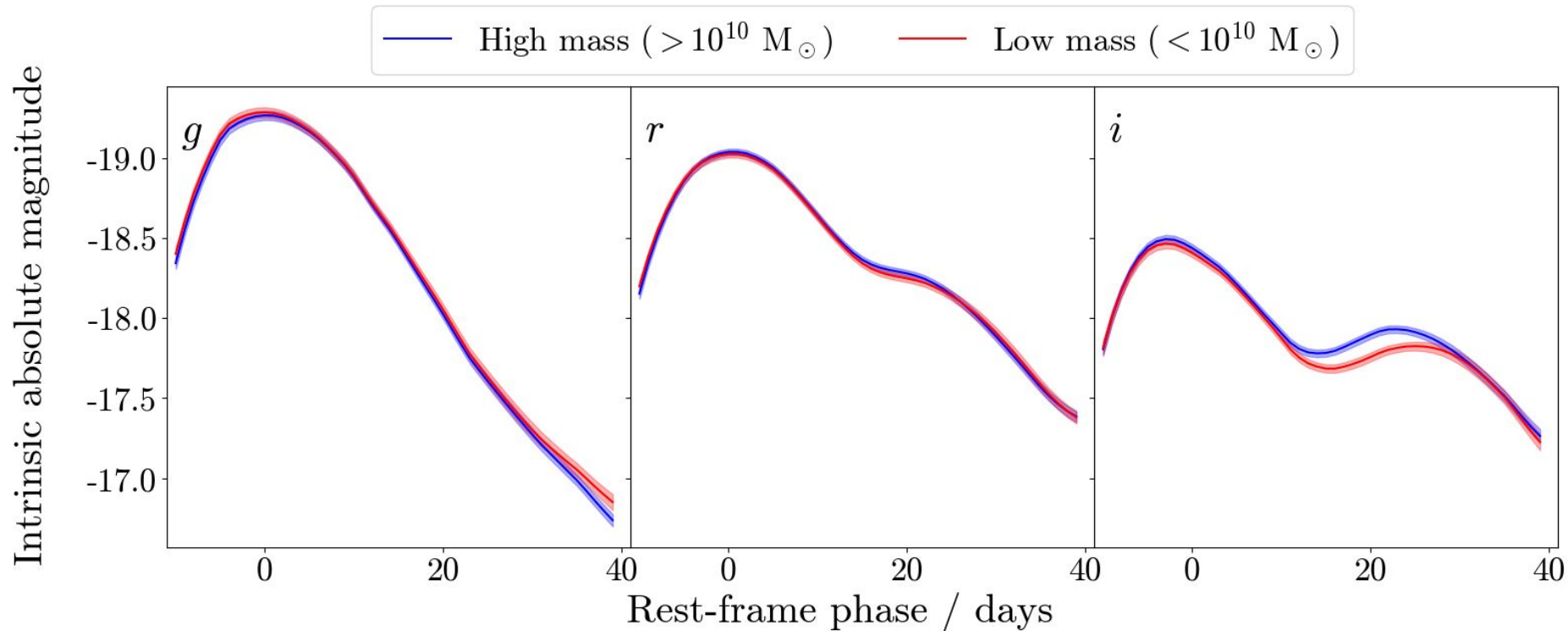
Dust may play a role, but BayeSN supports an intrinsic contribution to the mass step



Grayling & Popovic (2024)



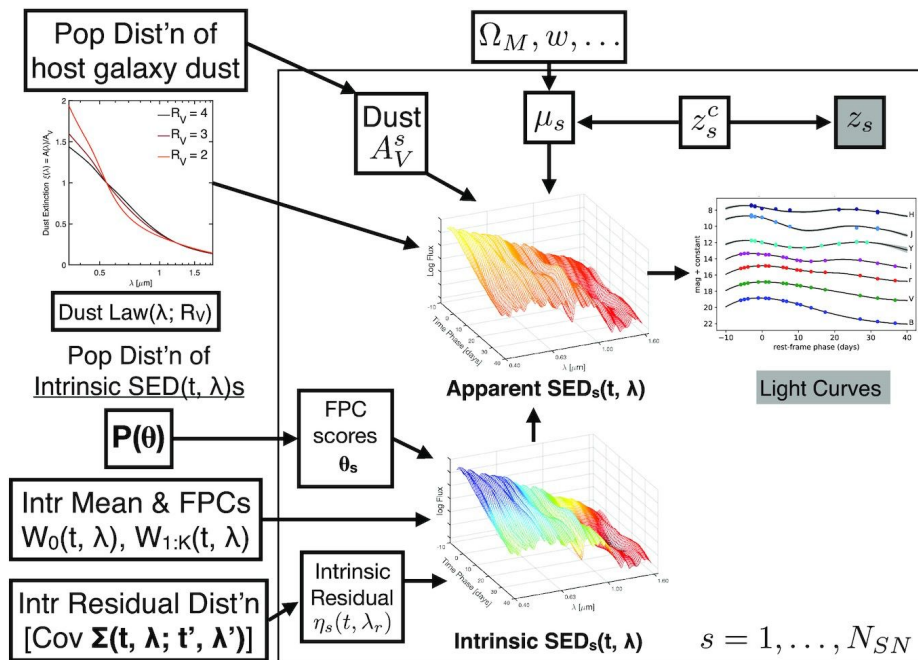
# DES5YR



Grayling & Popovic (2024)

# Impact of Supernova Astrophysics on Cosmology

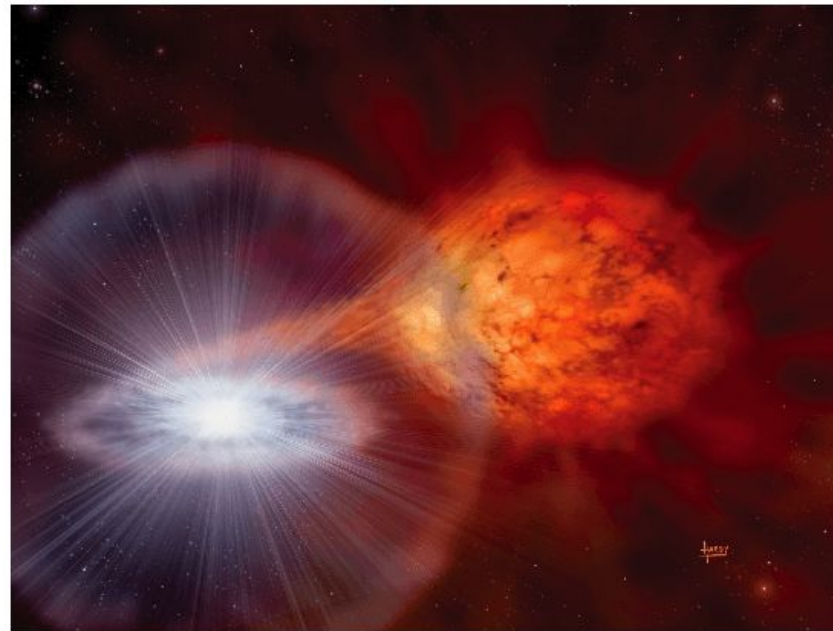
- We are reliant on empirical constraints on properties of SNe Ia
- Current analyses are step-by-step; first work out forward model, then correct for biases
  - These don't reflect our findings about intrinsic differences
- Ideally, we would infer cosmological parameters within BayeSN...
- ...but writing down the likelihood is impossible



Mandel+2020

# Talk Overview

- 1) Introduction to type Ia supernovae and supernova cosmology
- 2) Investigating the 'mass step' of type Ia supernovae
- 3) SBI for supernova cosmology

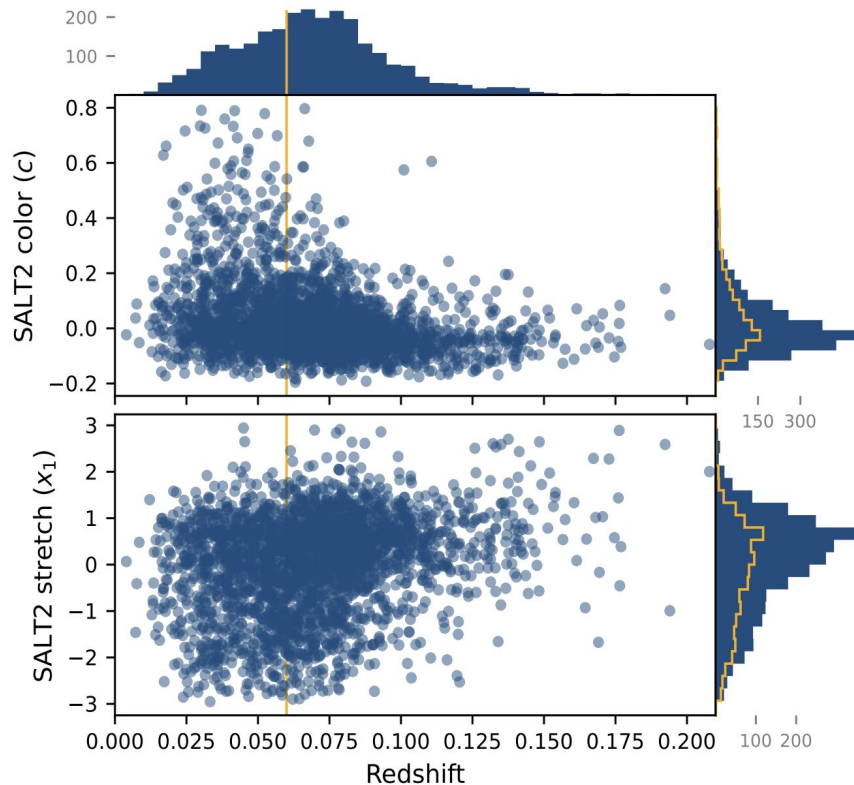


Credit: David A. Hardy, © David A. Hardy/astroart.org

# The challenge of likelihood-based inference

- Survey selection effects
  - Weather, number of observations, host redshift
- Contamination
  - Photometric samples will not be 100% pure, need to account for contaminants

We can simulate these effects, but expressing them analytically is impossible



# Simulation-based Inference (SBI)

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

- We want to do Bayesian inference, but the likelihood cannot be expressed analytically
- The solution:
  - Learn the likelihood/posterior from simulations!
  - Neural networks are excellent function approximators!

# Neural Ratio Estimation

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} \equiv r(\theta, D)P(\theta)$$

Neural Ratio Estimation

# Neural Ratio Estimation



Black cat



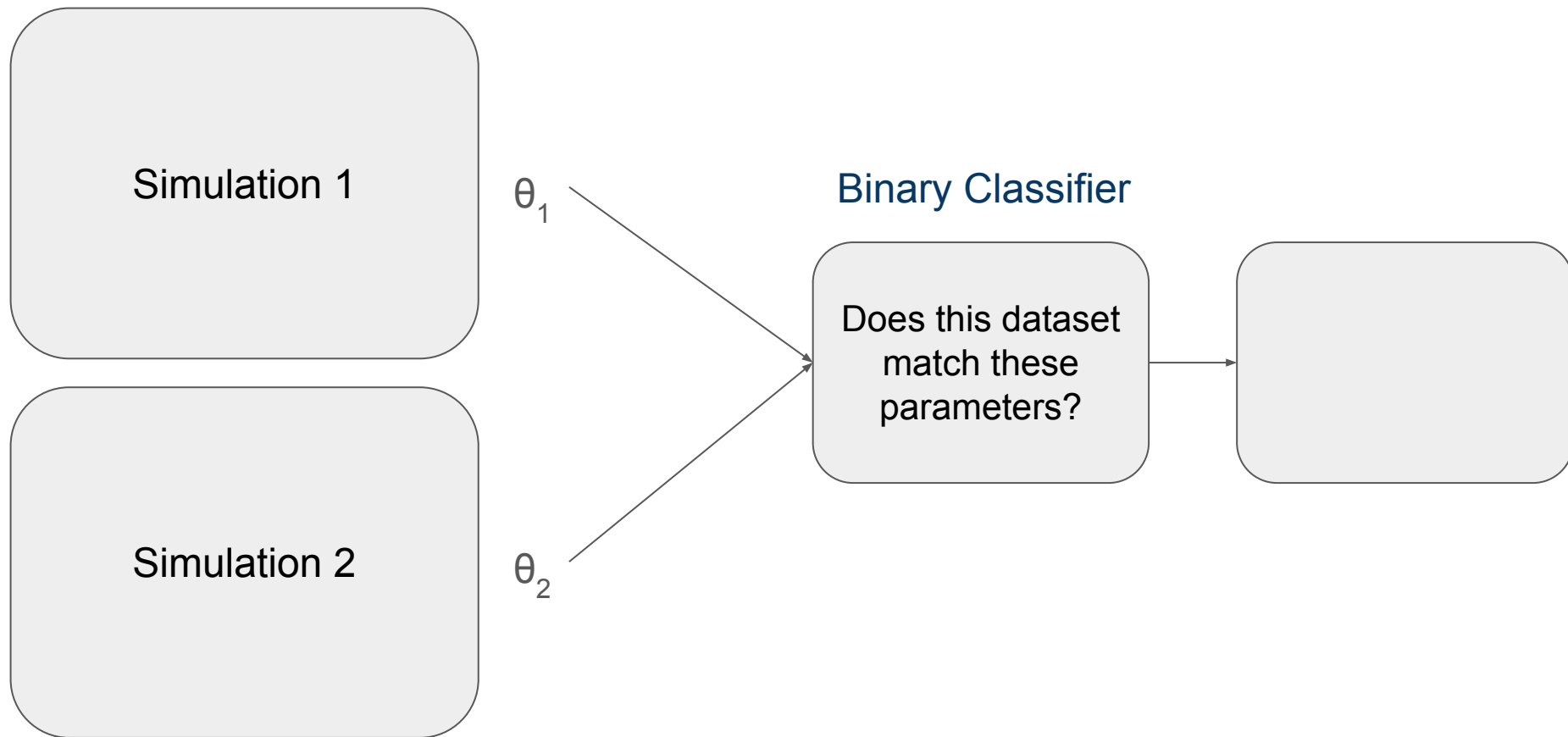
Ginger cat

Binary Classifier

Black cat or  
ginger cat?

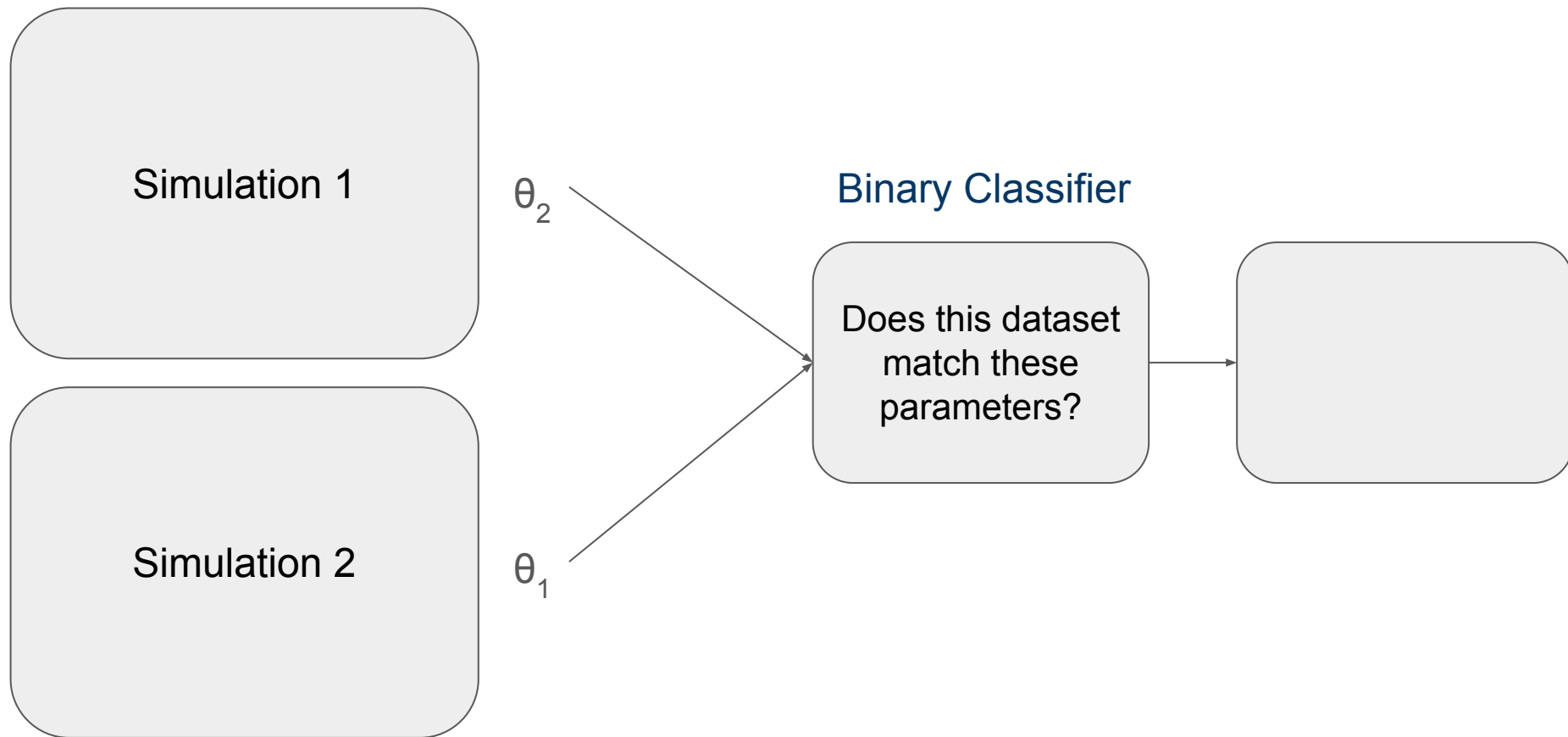
Output: 0 or 1

# Neural Ratio Estimation

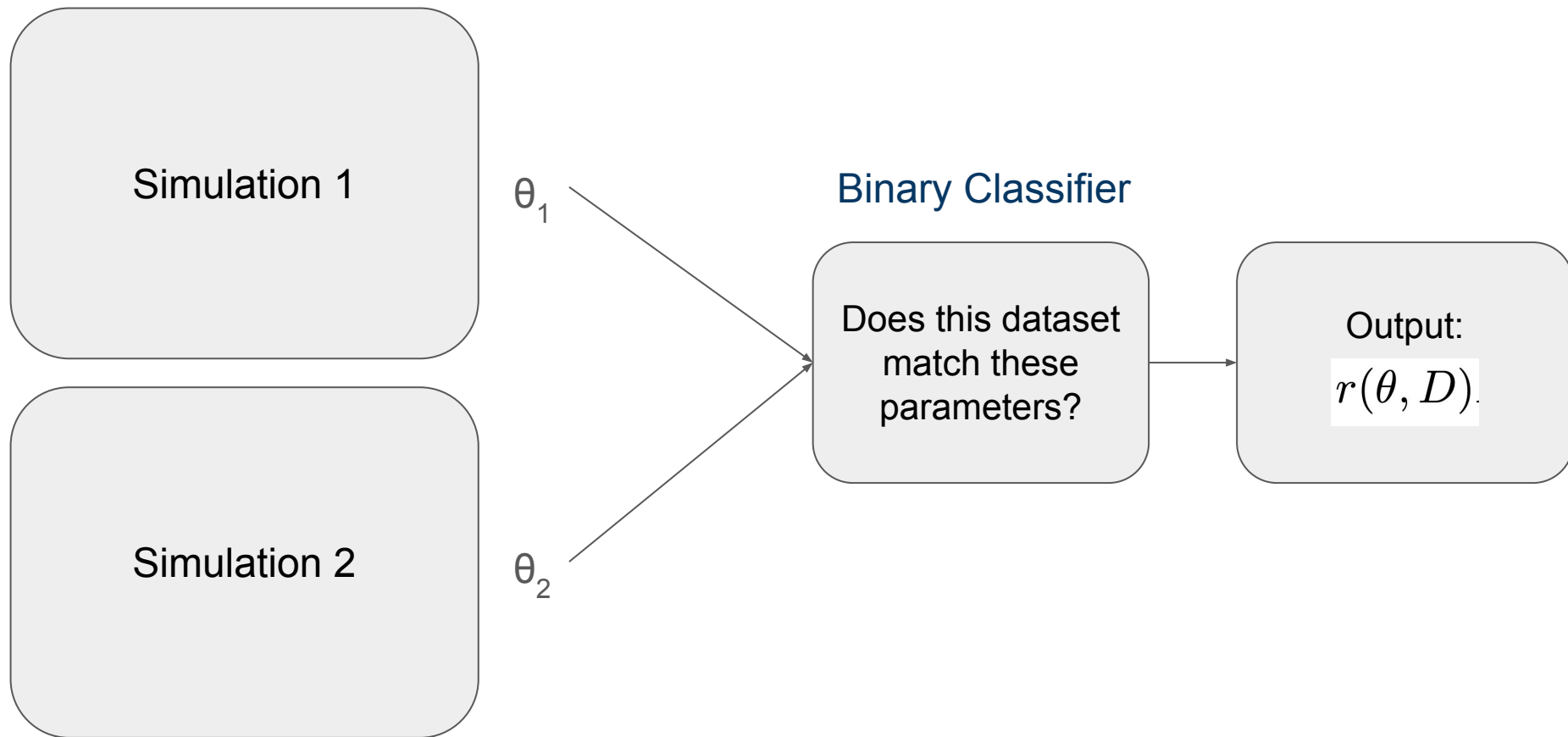




# Neural Ratio Estimation



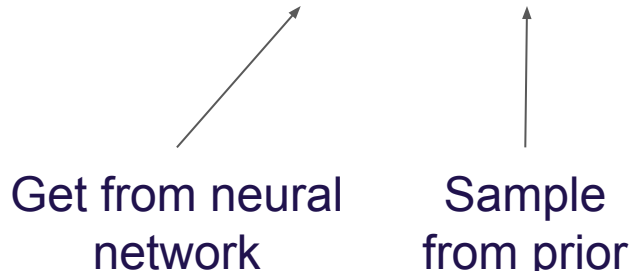
# Neural Ratio Estimation



# Neural Ratio Estimation

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} \equiv r(\theta, D)P(\theta)$$

Get from neural  
network

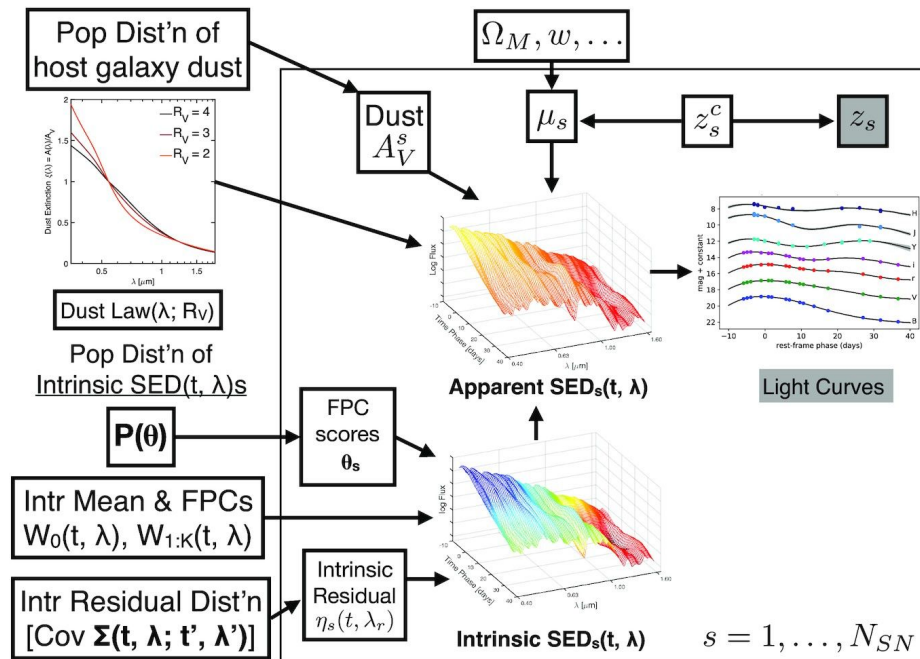


Sample  
from prior

- Gives you a posterior without ever calculating a likelihood!
- Other methods approximate different parts of Bayes' theorem

# The Future of Supernova Cosmology

- Single-step cosmological inference
- Cosmological parameters properly marginalised over empirical constraints on e.g. dust extinction, environmental dependence
  - Given the astrophysical uncertainty, this is vital!
- We can simulate lots of complex effects
  - Selection effects, photometric classification, photometric redshifts



Mandel+2020

# Strengths/Challenges of SBI

## Strengths:

- Easy to incorporate intractable, arbitrary effects
- Fast, amortised inference

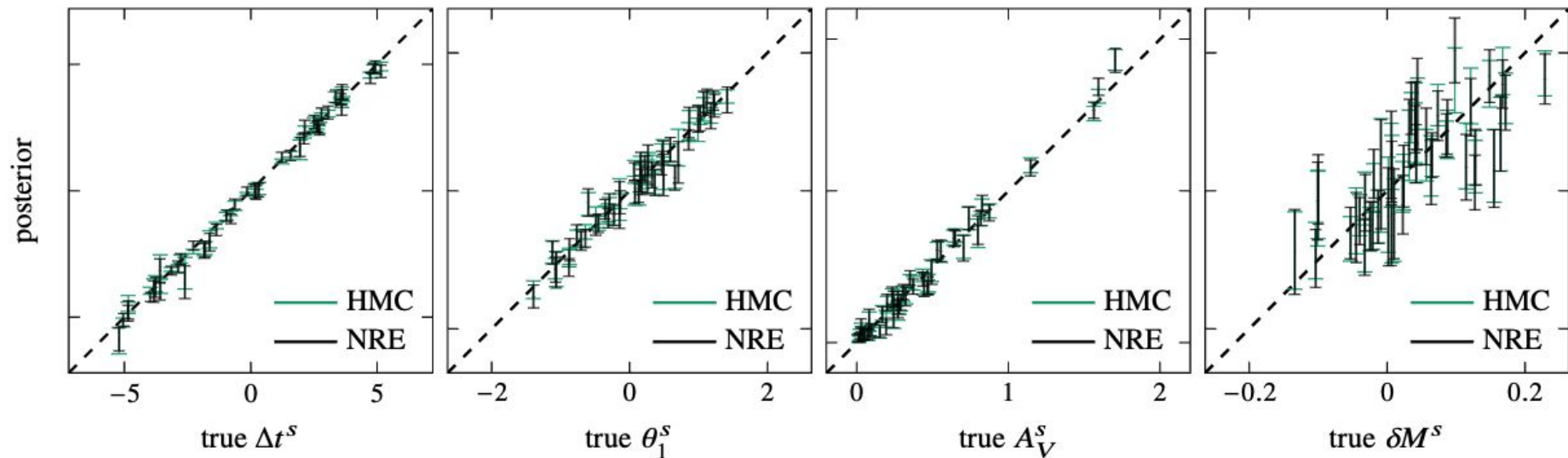
## Challenges:

- Need lots of simulations (SNANA not developed with SBI in mind)
- Variable-length sequences pose a challenge

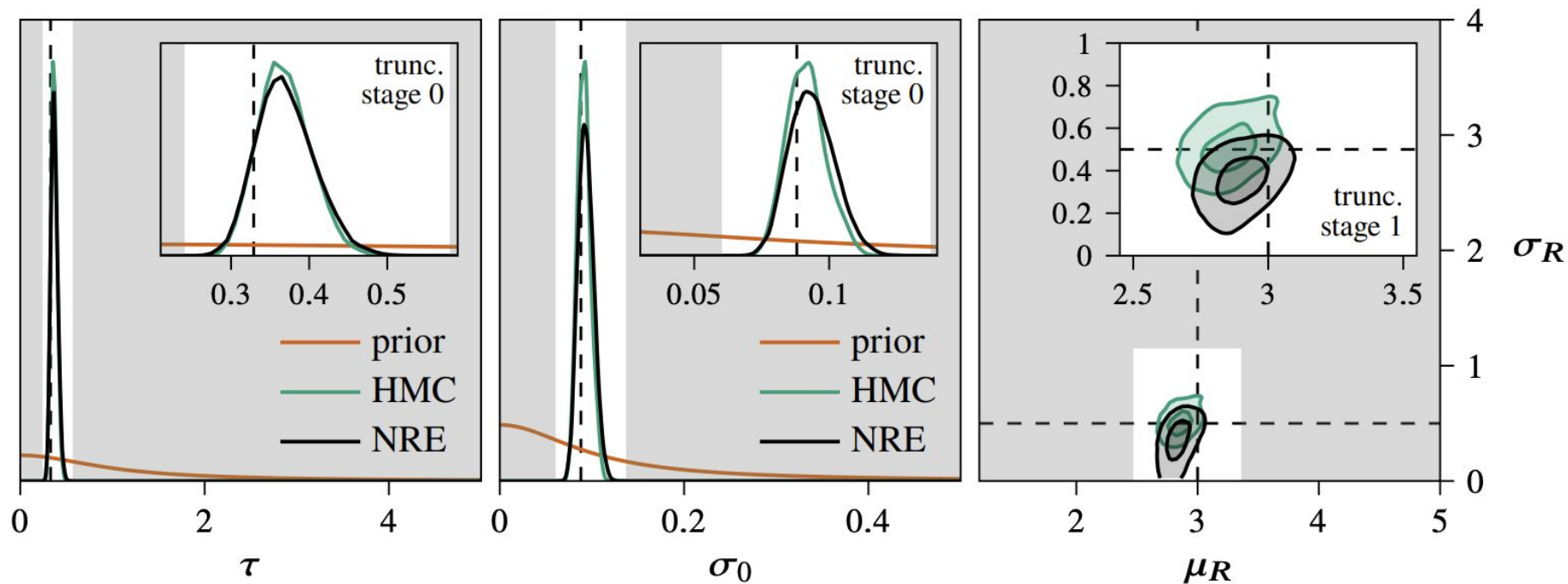
# Progress So Far

- Karchev, Grayling et al. 2024 recently applied truncated marginal neural ratio estimation (TMNRE) to dust inference of type Ia supernovae
  - Intermediate analysis was designed to allow for a comparison between TMNRE and HMC

# TMNRE for Dust Inference

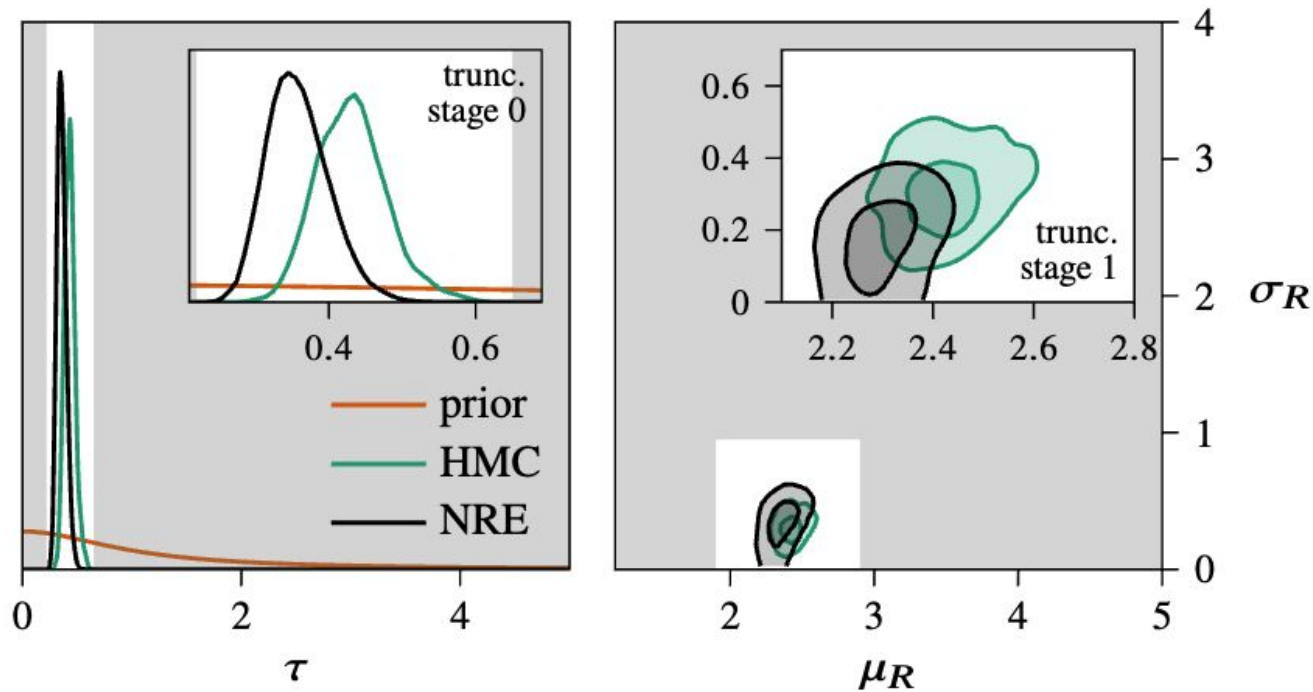


# Applied to Simulations





# Applied to Real Data



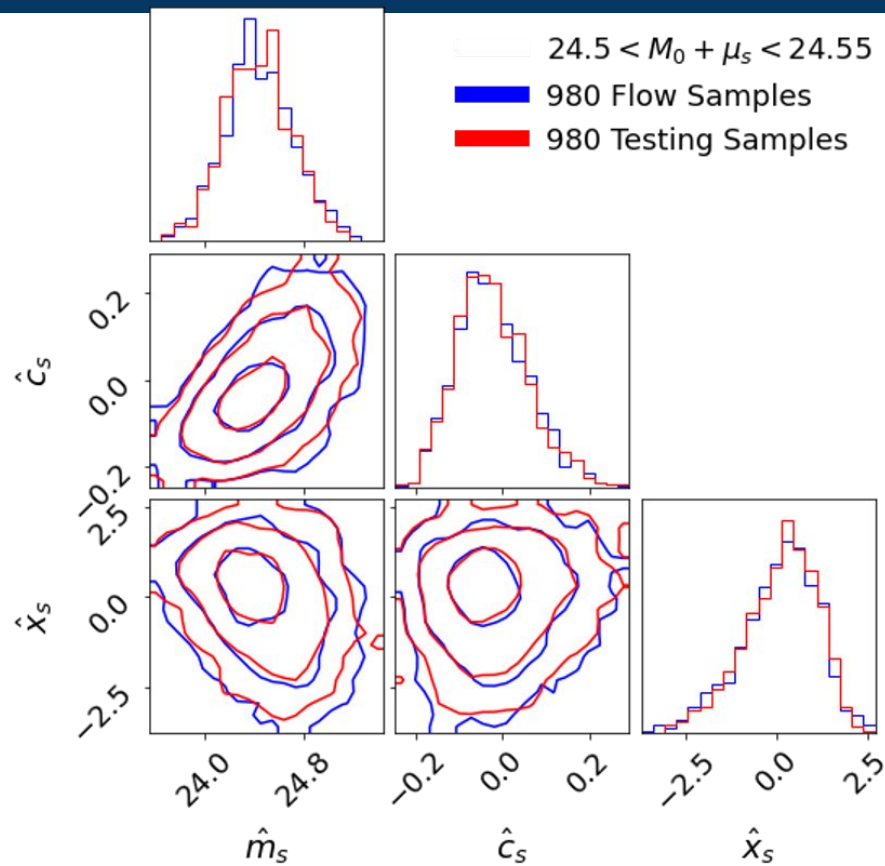
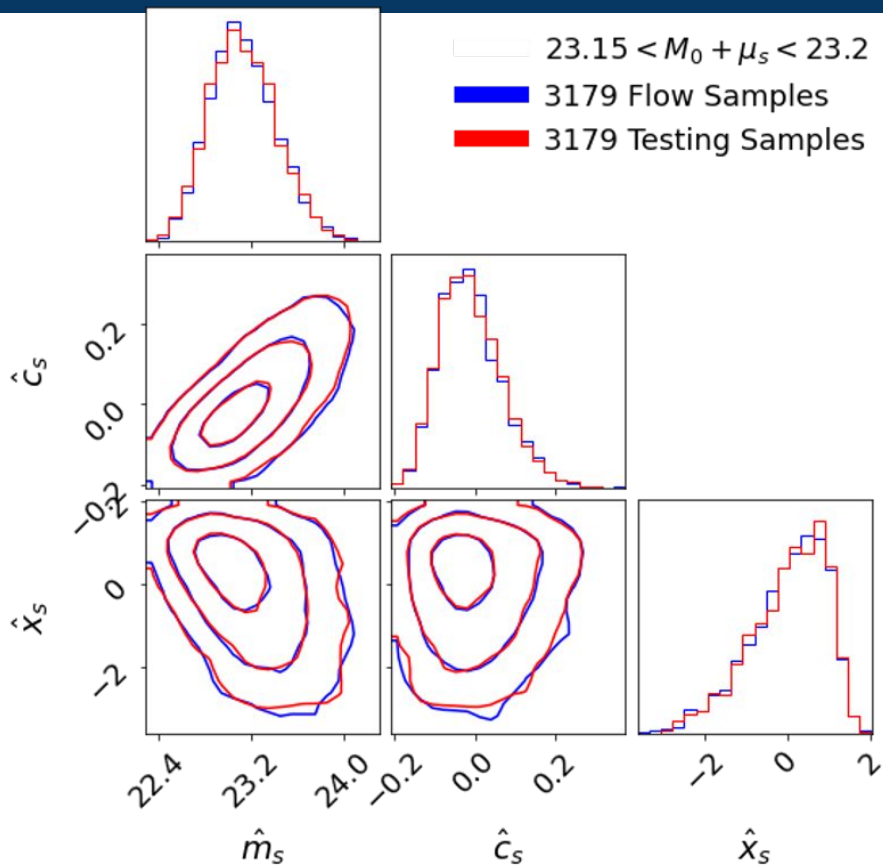
# Progress So Far

- Karchev, Grayling et al. 2024 recently applied truncated marginal neural ratio estimation (TMNRE) to dust inference of type Ia supernovae
  - Intermediate analysis was designed to allow for a comparison between TMNRE and HMC
  - Promising results, first application of NRE to real SN Ia data

# Progress So Far

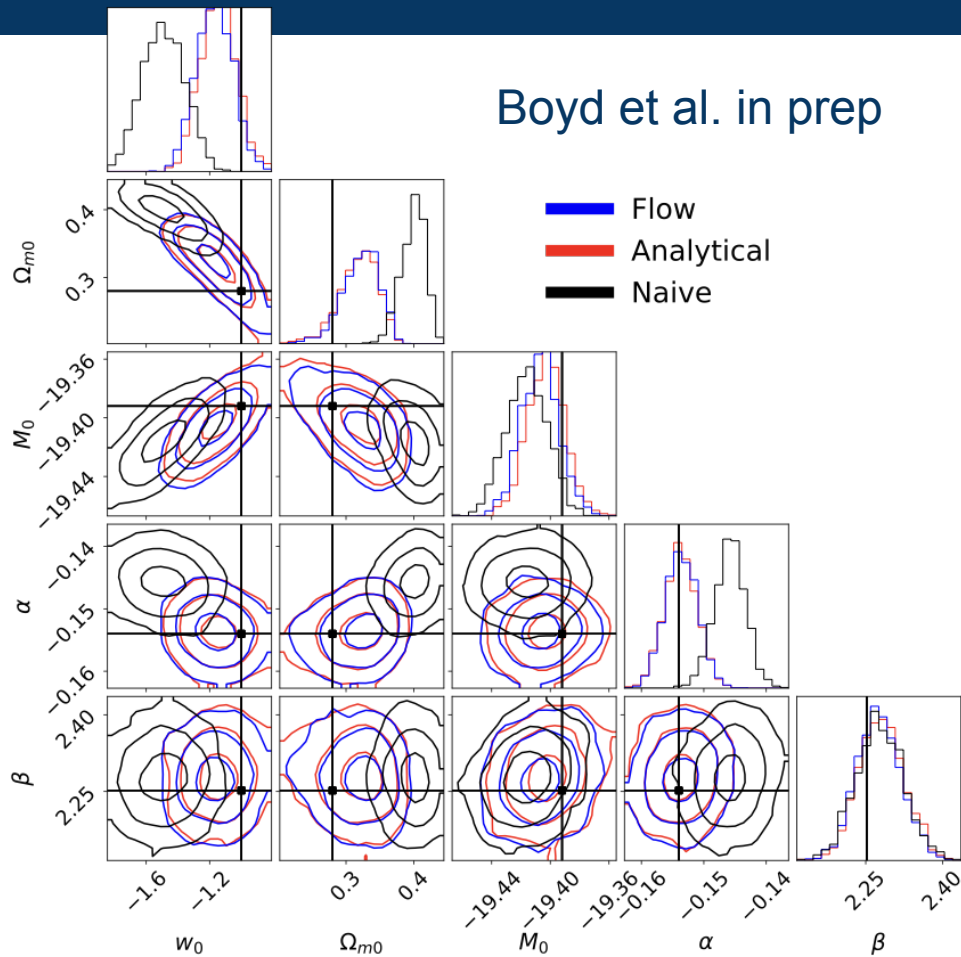
- Karchev, Grayling et al. 2024 recently applied truncated marginal neural ratio estimation (TMNRE) to dust inference of type Ia supernovae
  - Intermediate analysis was designed to allow for a comparison between TMNRE and HMC
  - Promising results, first application of NRE to real SN Ia data
- Boyd et al. 2024 applied normalising flows to learn likelihood incorporating selection function from simulations
  - Hybrid SBI approach, learn non-analytic likelihood to incorporate with typical MCMC inference

# Modelling Selection Function with Normalising Flows



# Modelling Selection Function with Normalising Flows

- Very closely matches truth with analytic selection function
- Also performing very well on more realistic survey simulations
- Embedding a neural network-based likelihood within MCMC performs well, maintains interpretability
- Paper to be submitted soon!



# Progress So Far

- Karchev, Grayling et al. 2024 recently applied truncated marginal neural ratio estimation (TMNRE) to dust inference of type Ia supernovae
  - Intermediate analysis was designed to allow for a comparison between TMNRE and HMC
  - Promising results, first application of NRE to real SN Ia data
- Boyd et al. 2024 applied normalising flows to learn likelihood incorporating selection function from simulations
  - Hybrid SBI approach, learn non-analytic likelihood to incorporate with typical MCMC inference
  - Very promising results

# The Road to SBI Supernova Cosmology

- There's a long road to reach real data SBI cosmology analysis, but we have made good progress
- Current work focusing on applying SBI to ever more realistic simulations, working towards real data
- It would be great to chat about SBI and fast sampling methods e.g. new HMC methods, vectorised samplers



# Summary

- With BayeSN, we find evidence to support an intrinsic component of the mass step though dust may also play a role
- We find a significant environmental-dependence of the intrinsic properties of SNe Ia, specifically on the i-band secondary maximum
- It's vital that cosmological analyses properly incorporate our empirical constraints on these effects.
- SBI will enable this, lots of work in progress moving towards a new future for SN cosmology



MNRAS paper



New arxiv paper